

Q2 2025 UPDATE

TowerXchange's European guide

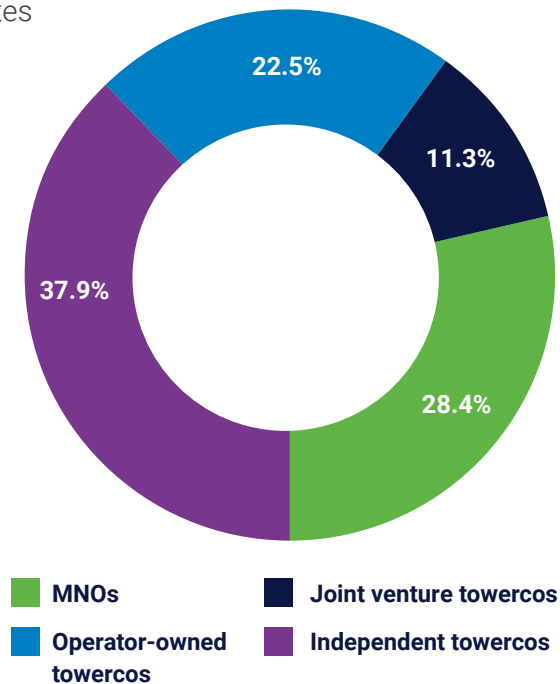
A country-by-country guide to the
European tower industry

Introduction

The closing of Swisscom's acquisition of Vodafone Italia in the beginning of 2025 not only reshapes the Italian market but also marks a significant change in the European telecommunications landscape with a few implications for mobile operators and towercos alike. It reduces the number of MVNOs in the country which could lead to further mergers or partnerships and strengthens its position in mobile and fixed services through increased market share. Integrating Fastweb's broadband infrastructure with Vodafone's mobile network will create synergies in infrastructure. This could also lead to potential future mergers on the continent as it sets a precedent for other MNOs to seek similar deals to remain competitive, and in turn affect more towercos. Vodafone Italia co-locates on INWIT and Cellnex towers which could trigger new contract negotiations by the new owner, or lead to a shift in tower strategies and rationalisation of existing sites.

In Spain, Vodafone is reportedly working with MasOrange to merge mobile networks and set up a joint venture. The new JV will create synergies in leasing space on towers, equipment, transmission and workforce. No figures on expected savings have been disclosed yet. Vodafone and MasOrange already have an active sharing agreement in place in Spain set up before the merger of MasMovil and Orange, which is in areas of the country where it's not economically viable to have two networks. One recent example from Spain is Vodafone Spain's new owner

Figure 1: Ownership of Europe's estimated 747,875 sites



*Figures include deals that have closed only

Source: TowerXchange

Zegona Communications attempt to negotiate an EUR 50mn reduction in rent on leasing space on Vantage Towers where it is the anchor tenant.

In the UK, Vodafone received regulatory approval to

merge with Hutchison's Three which will bring another shake-up for competition, and is expected to bring better infrastructure investments, improving 5G coverage and mobile network efficiency. The final outcome will depend on the conditions and remedies the UK government imposes.

MNO consolidation has also taken place in Hungary. Vodafone Hungary and DIGI have consolidated into One to form a single operating entity from January 2025. Both MNOs were acquired by parent company 4iG from former owners Vodafone in 2023 and in 2022 from Romanian Digi Communications respectively. The move has created the second largest telecommunications company in the country and is also part of a larger business transformation journey for the group.

Keeping east, Hrvatski Telekom has announced the start of the process of separating 1,700 of its macro towers into a wholly owned subsidiary, expecting to improve operational efficiency and bottom line. The process is pending corporate approvals and is expected to be completed in the third quarter of 2025.

TowerXchange understands that MNO Tele2 is considering selling its towers in the three Baltic countries and is working with Perella Weinberg Partners to reportedly try to fetch 500mn Euros for them. Meanwhile, the sale of Bite's towers in Latvia and Lithuania is still ongoing.

INTRODUCTION

Alongside the ongoing M&A activity and consolidation across the continent, 5G (and in a few countries 4G) rollout continues to present significant co-location and new build opportunities for Europe’s towercos. In addition, street works are increasingly being used to deliver macro-cell like capabilities with small cell like footprints, and outside of the urban centres more joint ventures are being formed to address rural coverage.

Alongside more traditional revenue streams, 5G is further accelerating towercos’ focus into new service lines. The provision of in-building solutions is now commonplace for many towercos. The outdoor small cell coverage requirements spurred by the densification needs of MNOs

are creating new revenue streams. In addition, public transport network coverage has seen particular growth as EU initiatives are channelling more focus into transport networks presenting monetisation opportunities.

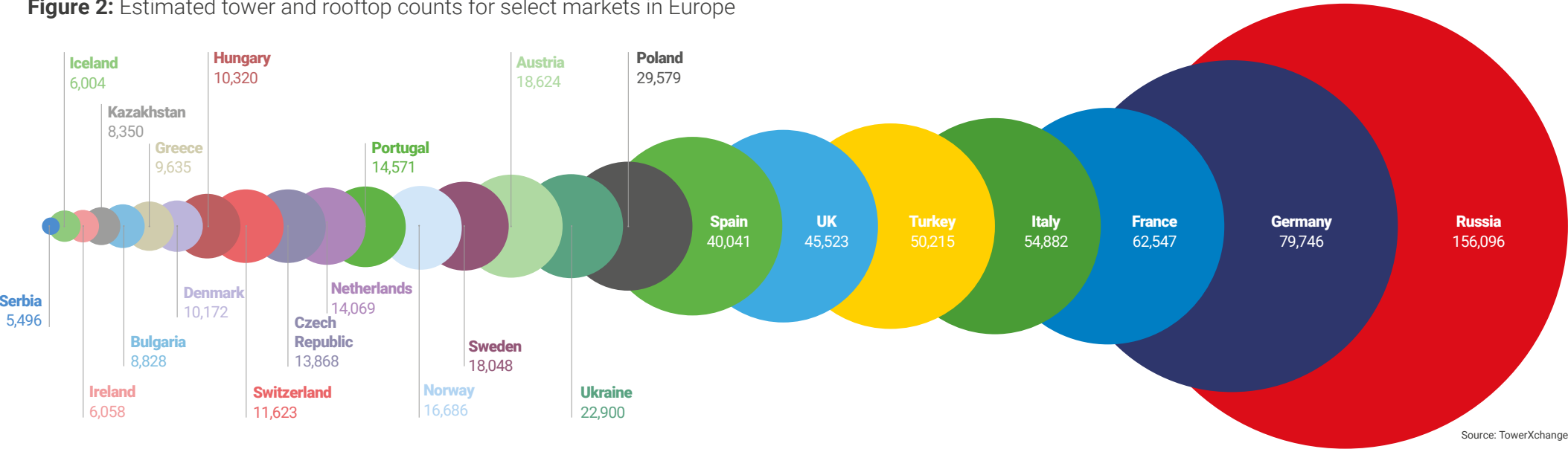
Plus, whilst traditionally associated with towercos operating in developing markets rather than those in a good grid market such as Europe, the provision of energy and energy services is starting to feature more significantly in towercos’ service offerings. Escalating energy consumption by networks, increased electricity costs and a heightened focus on ESG are all driving further emphasis on power and power security. Exploring clean energy solutions has become a strategic direction for European towercos with

some already reporting onsite renewable energy generation.

While Cellnex in Poland and CETIN in the Czech Republic are known as the towercos managing active infrastructure, it appears a few other towercos are exploring RAN management. As 5G densification in urban areas advances, capex-restricted MNOs are expected to consolidate some networks and arrive at synergies and efficiencies.

At TowerXchange Meetup Europe 2025, we’ll explore Towerco 2.0 and the new roles towercos embrace at several sessions across the agenda. Join us in London to meet established and new European towercos, hear the latest on financing, policy and commercial opportunities, and make new connections.

Figure 2: Estimated tower and rooftop counts for select markets in Europe



**Figure 3:** Tower counts of Europe's towercos

* Arqiva 900 / České Radiokomunikace 800 / EI Towers 700 / Towercom (Slovakia) 700 / Emitel 667 / Axion 600 / HighTel Towers 600 / Mila 500

* Cellnex numbers as reported by the company, do not include subsequent changes in UK

[i] Vantage Towers' site count includes their stakes in INWIT and Cornerstone. INWIT and Cornerstone are also separately listed.

[ii] Telenor Tower Holding's site count excludes their stakes in 3GIS, Net4Mobility & TT-Network

[iii] Representing all Telia's towerco operations combined

Figure 4: Tower strategies of Europe's major MNOs

Company	Other alias'	European countries of operation	Tower strategy
A1 Group (Telekom Austria)	A1, vip	Austria, Belarus, Bulgaria, Croatia, North Macedonia, Serbia, Slovenia	Telekom Austria Group have carved out their towers in Austria. The Group's assets there, together with the ones in Bulgaria, Croatia, North Macedonia, Slovenia and Serbia will be managed under the EuroTeleSites towerco which has been headquartered in Vienna and listed on the Vienna Stock Exchange. The Belarus assets are out of scope.
Altice	SFR	France; Portugal	In France, carved out their towers into a captive towerco, Hivory, in which they sold a 49.99% stake to KKR. Hivory has subsequently been sold to Cellnex. In Portugal, carved their towers into OMTEL, with OMTEL since 100% acquired by Cellnex
Deutsche Telekom	Telekom Deutschland; T-Mobile; Magenta Telekom	Austria, Czech Republic, Germany, Netherlands, Poland, Slovakia (Plus direct and indirect investments in Croatia, Greece, Hungary, Montenegro and North Macedonia, Romania)	German towers carved into Deutsche Funkturm in 2002; Austrian towers carved into Magenta T-Infra in 2021. Both are held under the GD Towers business unit. In July 2022, Deutsche Telekom agreed the sale of a 51% stake in GD Towers to DigitalBridge and Brookfield. Dutch towers have been merged with Cellnex' Netherlands' portfolio, with Cellnex and Deutsche Telekom creating a new investment fund, Digital Infrastructure Vehicle which holds a 38% stake of Cellnex Netherlands as its seed asset. DIV plans to seek third party investment and invest in other digital infrastructure. In Poland, T-Mobile's towers are managed through a JV with Orange (NetWorks!) Deutsche Telekom has carved out its towers (macros and rooftops) in the Czech Republic into T-Mobile Infra CZ and in Slovakia into Slovak Telekom Infra. It has announced plans to spin off its Greek assets.
Hutchison	3; Hi3G; Wind Tre	Austria, Denmark, Ireland, Italy, Sweden, UK	Towers sold to Cellnex for €8.6bn in cash and €1.4bn in Cellnex shares (c. 5% Cellnex' capital). Hutchison's UK towers are managed by the JV MBNL, with Cellnex set to acquire only the economic risks and rewards of the towers. In Sweden, a portion of Hutchison's towers are managed by the JV, 3GIS. The UK portion of Hutchison's deal closed in Q4 2022 with the acquisition of 1,100 towers by Wireless Infrastructure Group from Cellnex. In June 2023 Vodafone and Three announced their long-anticipated merger. Pending regulatory approval, Vodafone will own 51% of the combined business (Mergeco) with Three's parent company CK Hutchinson owning the remaining 49%. In Italy, the sale of Hutchison-owned Wind Tre's carved out fixed and mobile network assets to EQT Infrastructure has been cancelled due to 5G agreements with third parties.
Iliad (and other MNOs owned by Xavier Niel's NJJ)	eir, Epic, Free, Iliad, Monaco Telecom, Salt	Cyprus, France, Ireland, Italy, Malta, Monaco, Poland	Xavier Niel owned companies have sold their towers to Cellnex in four markets. In 2019, Iliad sold their towers in France and Italy to Cellnex, retaining a 30% stake in the former (Cellnex having since acquired the remaining stake). At the same time, Salt sold their towers to the towerco, retaining a 10% stake. In Poland Cellnex has bought out Iliad's 30% stake in the JV OnTower Poland. In Ireland, eir has sold their tower portfolio to Phoenix Tower International and the towerco has also acquired Monaco Telecom's towers in Malta and Cyprus. In France Iliad has signed a further network sharing deal with TOTEM.

**Figure 4:** Tower strategies of Europe's major MNOs

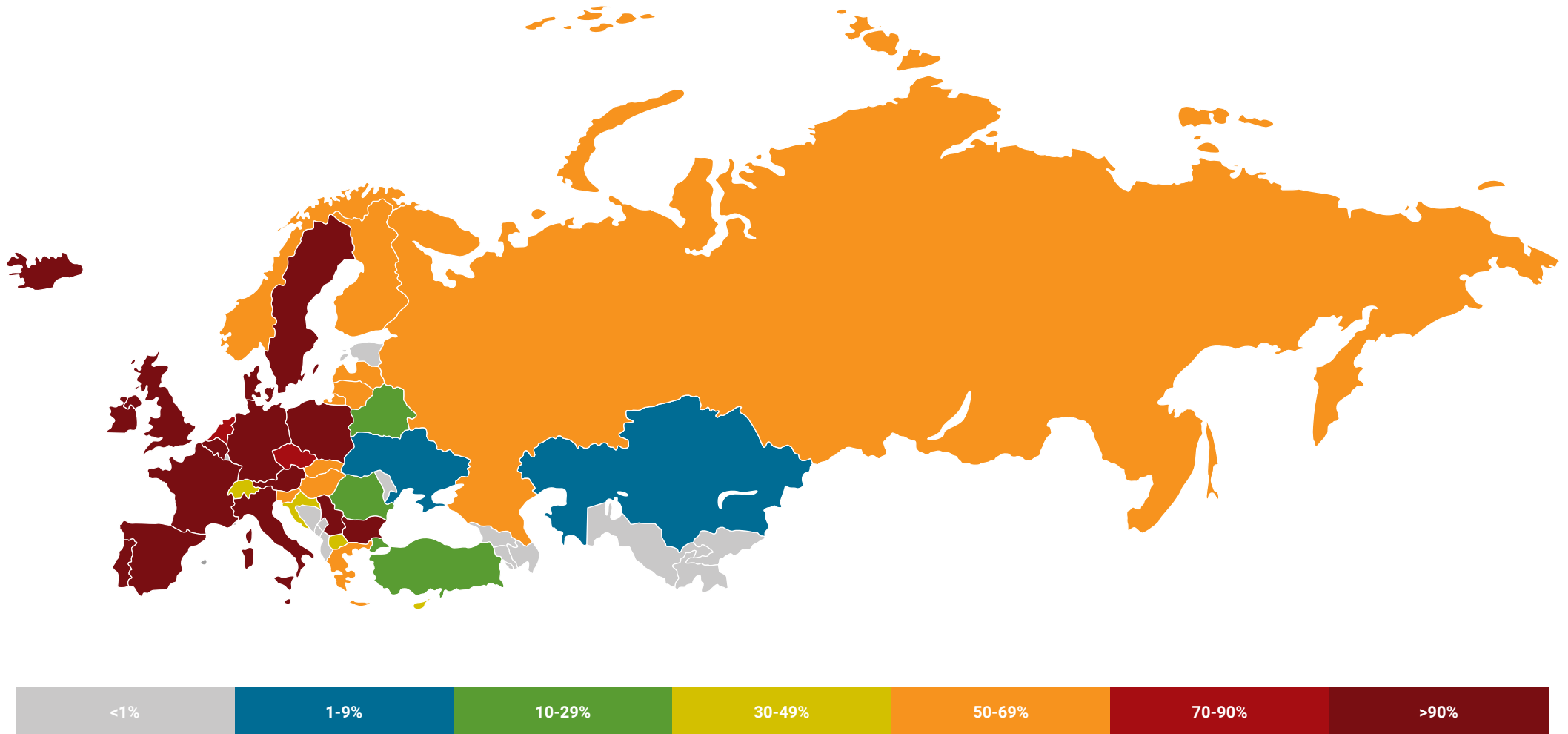
Company	Other alias'	European countries of operation	Tower strategy
Liberty Global	Sunrise, Telenet, Virgin Media O2, VodafoneZiggo	Belgium, Netherlands, Switzerland, UK	Liberty Global has stated that it is exploring options regarding monetisation of its assets. In Belgium, Liberty Global's opco, Telenet, sold its towers to DigitalBridge's Belgium Tower Partners. In Switzerland, Sunrise sold its towers to Cellnex prior to the opco's takeover by Liberty Global. In the UK, Virgin Media O2 (Liberty Global's JV with Telefónica) sold a 16.67% stake in Cornerstone to GLIL Infrastructure for GBP360mn (US\$459mn).
Orange		Belgium, France, Luxembourg, Moldova, Poland, Romania, Slovakia, Spain	Orange has carved out its French and Spanish towers into TOTEM which launched in October 2021. The operator is widely expected to bring tower portfolios from its other markets into TOTEM's perimeter and is undertaking a strategic review at present. Orange wants to remain majority control of TOTEM. TOTEM has an appetite for M&A including the potential formation of JVs with other MNO owned towercos. In Poland and Romania, Orange's towers are managed by the joint ventures NetWorkS! (with T-Mobile) and Netgrid Telecoms (with Vodafone) respectively, although the towers remain on the balance sheets of the MNOs. Outside of this, Orange has completed one sale and leaseback transaction in Europe, selling 1,500 non-strategic sites to Cellnex in Spain.
PPF Telecom Group	O2 Czech Republic, Yettel	Bulgaria, Czech Republic, Hungary, Serbia	PPF has separated out its infrastructure and retail businesses in the Czech Republic, Bulgaria, Hungary and Serbia with the infrastructure managed under CETIN Group and in Slovakia under O2 Networks. CETIN manages both passive and active infrastructure. In Q4 2021, PPF announced the sale of a 30% stake in CETIN to Singapore's GIC investment fund. It also sold its 100% stake in Montenegro to Hungary's 4IG. In 2023 it announced the sale of 50% +one share in its MNO ad towerco units only in Bulgaria, Hungary, Serbia and Slovakia.
Tele2		Estonia, Kazakhstan, Latvia, Lithuania, Sweden	Has yet to reveal any significant plans for their passive infrastructure but Tele2 told analysts in their Q320 results call that "there is an opportunity to do more in terms of sharing and separating out some of our infrastructure assets". The company has not made public statements since. In Sweden, Tele2 is involved in two active sharing joint ventures; SUNAB (with Telia, covering 3G which is in the process of being wound down) and Net4Mobility (with Telenor – whose remit has just been extended to include 5G as well as 2G and 4G). In Latvia and Lithuania, Tele2 had formed Centuria, a network sharing joint venture with Bite but the joint venture was wound down and did not become operational after local opposition.
Telefónica	Movistar, Virgin Media O2	Germany, Spain, UK	Telefónica carved their infrastructure assets (both towers and subsea cabling) into infraco Telxius before selling a 40% stake to KKR and an effective 9.99% stake to Pontegadea. Telefónica subsequently sold Telxius' tower business to American Tower. In the UK, Telefónica's towers are owned by Cornerstone, Telefónica's 50:50 joint venture with Vodafone. Telefónica and Vodafone recently announced the commercialisation of Cornerstone, with Vodafone transferring its stake to Vantage Towers. Virgin Media O2 (Liberty Global's JV with Telefónica) sold a 16.67% stake in Cornerstone to GLIL Infrastructure for GBP360mn (US\$459mn).

**Figure 4:** Tower strategies of Europe's major MNOs

Company	Other alias'	European countries of operation	Tower strategy
Telenor		Denmark, Finland, Norway, Sweden	Telenor has carved out their Nordic tower assets into Telenor Tower Holdings, focussing on increasing the external revenue and improving operational efficiencies. From Q1 2022, Telenor been reporting financials for the assets, although their stakes in various joint ventures have been excluded from this reporting. In Sweden, Telenor is involved in two active sharing joint ventures: 3GIS (with Hi3G which covers 3G and is being wound down) and Net4Mobility (with Tele2 – whose remit has just been extended to include 5G as well as 2G and 4G). In Denmark, Telenor's towers are held by TT-Network, the joint venture with Telia. Telenor plans to monetise a minority stake in one or more of its national towercos in the next 12-24 months.
Telia	Moldcell	Denmark, Estonia, Finland, Latvia, Lithuania, Moldova, Norway, Sweden	Telia's tower assets in Sweden, Norway and Finland have been carved into separate towerco units and sit under the Telia Asset Management business unit. In June 2021, Telia agreed the sale of a 49% stake in its Finnish and Norwegian assets to infrastructure investor Brookfield and Swedish pension fund, Alecta. In January 2022, a similar deal was agreed in Sweden. The operator is involved in a number of infrastructure sharing joint ventures. In Denmark Telia has announced it's sold its business to energy-telco group Norlys for an enterprise value of DKK 6.25bn (USD\$882.48mn) including its joint venture with Telenor, TT-Network. In Sweden, Telia's infrastructure sharing joint venture with Tele2 covers just 3G and is set to be unwound as the technology is sunsetted
Turkcell	BEST	Belarus, Turkey, Turkish Republic of Northern Cyprus	Turkcell has carved out its towers across its operations into Global Tower. In September 2024 it sold Ukrainian assets – MNO lifecell and towerco UkrTOWER to DVL Telecom, a NJJ Holding Company.
United Group	Wind Hellas, Telemach, Vivacom	Bulgaria, Croatia, Greece, Slovenia	In August 2023 United Group completed the sale of its assets in Bulgaria, Croatia and Slovenia to Saudi towerco TAWAL. . In Greece, Wind's previous owners sold its tower portfolio to Vantage Towers prior to the opco's takeover by United Group. In addition to sharing passive infrastructure through Vantage Towers, Vodafone and Wind Hellas have an active sharing agreement in place.
VEON	Beeline; Kyivstar;	Kazakhstan, Kirgizstan, Russia, Ukraine, Uzbekistan	In Pakistan, the operator had already carved out its towers into Deodar and their sale to UAE-based TASC is currently on hold. In Bangladesh, carve out proceedings are well underway, with a sale process widely expected to be announced soon. Prior to the war erupting in Ukraine, VEON had also begun tower carve out proceedings in the country. VEON (then VimpelCom) had previously sold its Italian towers to Cellnex, prior to divesting its Italian operations. Veon exited Russia in Oct 2023 by closing the sale of its Russian operations to a group of senior members of the PJSC VimpelCom management team, led by VimpelCom CEO Alexander Torbakhov.
Vodafone		Albania, Czech Republic, Germany, Greece, Hungary, Ireland, Italy, Malta, Netherlands, Portugal, Romania, Spain, Turkey, UK	Vodafone sold its 81.7% stake in its pan-European towerco Vantage Towers and taken the publicly trading company private. Vantage Towers has an appetite for M&A and the formation of joint ventures. In Greece, Vodafone has acquired the towers of Wind Hellas; In Italy Vodafone has merged its towers with TIM's INWIT, with Vantage Towers owning a 33.2% stake in INWIT; In the UK, Vodafone has transferred its 50% interest in JV Cornerstone to Vantage Towers, whilst simultaneously announcing the commercialisation of Cornerstone. In Romania, Vantage Towers' sites are managed (but not owned) by Netgrid Telecoms, their joint venture with Orange.



Figure 5: Towerco penetration across the European continent



Austria

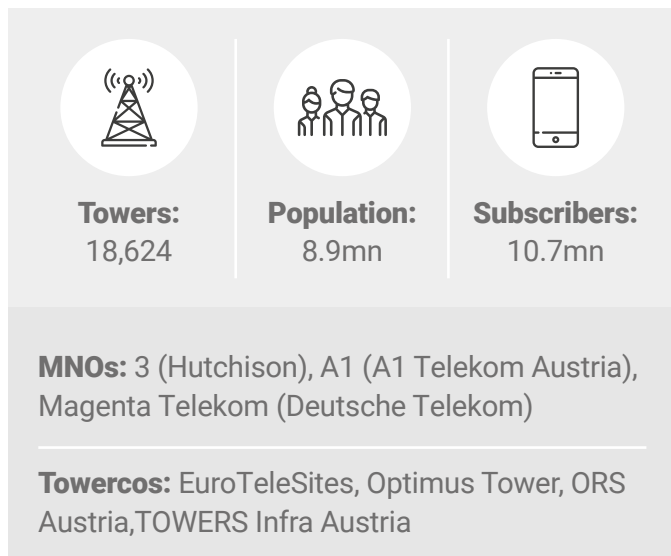
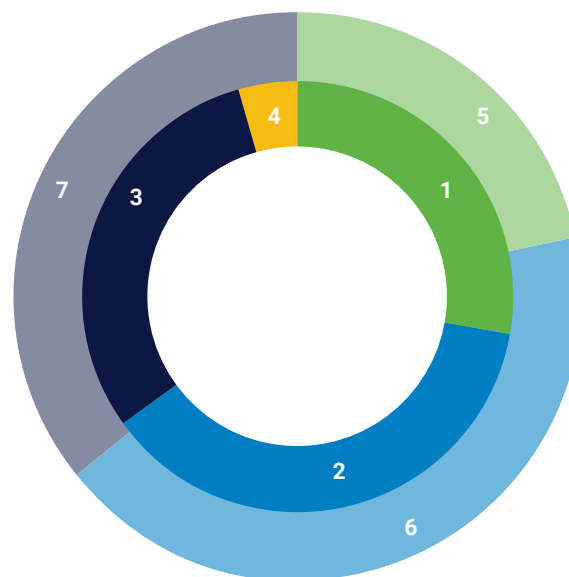


Figure 6: Austria – estimated tower ownership



Ground based towers

- 1. Optimus Tower 2,792
- 2. TOWERS Infra Austria 3,721
- 3. EuroTeleSites 3,067
- 4. ORS Austria 430

Rooftops

- 5. Optimus Tower 1,862
- 6. TOWERS Infra Austria 3,669
- 7. EuroTeleSites 3,067

Source: TowerXchange

Magenta Telekom (owned by Deutsche Telekom), had carved out its c. 7,000 towers into a separate business unit which was held alongside Deutsche Telekom's German towerco, Deutsche Funkturm, under the GD Towers brand. In July 2022, Deutsche Telekom reached a deal to sell a 51% stake in GD Towers to DigitalBridge and Brookfield. Deutsche Telekom retains a 49% stake in the entity along with significant majority protection rights, including the ability to appoint two of GD Towers' five shareholder committee members, including the Chairman.

A1 Telekom Austria Group has completed the carve out of its Austrian towers. The move follows the group's carve out of towers in Bulgaria, Serbia, Croatia, Slovenia and North Macedonia into separate legal entities (A1 Towers) between 2021 and 2022. The assets in the six countries will be managed by the Vienna-listed EuroTeleSites towerco.

EuroTeleSites has announced A1 will be the anchor tenant in its geographies and has set forth plans to focus on 5G upgrades and network densification in the short term, and on DAS, IBS and 5G rollout of new entrants and IoT businesses in the medium term.

The Austrian market opened up to the towerco model in late 2020/ early 2021 with Cellnex acquiring Hutchison's tower portfolio in the market as part of a [pan-European deal](#).

In August 2024, Cellnex sold its business to a consortium formed by Vauban Infrastructure Partners, through Core Infrastructure Fund IV SCSp and Core Infrastructure Fund IV SCA SICAV RAIF, EDF Invest, the investment arm of EDF for non-listed Dedicated Asset and MEAG, the asset manager of Munich Re and ERGO. The assets are now managed by Optimus Tower.

Belgium

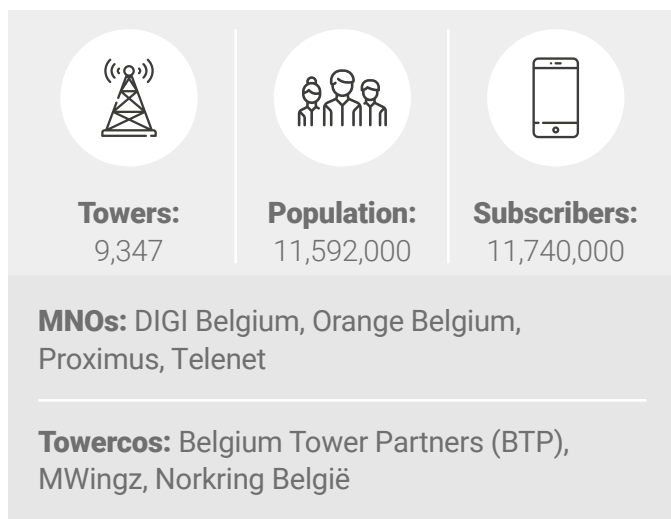
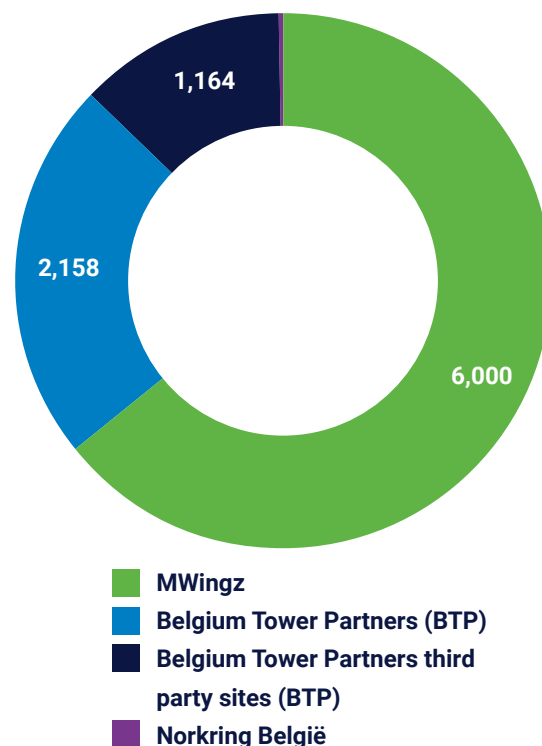


Figure 7: Belgium – estimated tower ownership



Source: TowerXchange

and has a 23% market share. Liberty Global increased its stake in Telenet to 100% through a simplified “squeeze-out” (enabling it to complete a voluntary takeover) in October 2023 and delisted the company. It has been an active

player in Belgium’s mobile market through an MVNO since 2006 (several other MVNOs continue to serve the Belgium market). In April 2015, Telenet acquired a Belgium MNO called Base from Dutch KPN, strengthening its position on the market and migrating its mobile customers to BASE’s network. Belgium’ first independent towerco was created in March 2022 when DigitalBridge acquired 3,322 sites from Telenet for €745mn. The towerco has a nation-wide footprint. Following the completion of the transaction DigitalBridge renamed the company Belgium Tower Partners (BTP). The deal includes 2,158 sites owned outright, as well as 1,164 third party sites over which BTP has less ability to lease up. The agreement includes a build-to-suit (BTS) commitment to deploy a minimum of 475 additional new sites, with Telenet acting as a subcontractor taking the lead on construction.

Romania-based multi-play service provider Digi entered the Belgium market in 2022 by setting up a joint venture with B2B telecom player Citymesh. The commercial launch of Belgium’s fourth MNO was on 11 December 2024 and has shaken up the market with its cheap services. Its scheduled major nationwide 5G mobile network deployment programme for 2024 seems to be behind schedule. Neither Digi Belgium nor its parent company Digi Communications have reported any results, and the MNO was not mentioned in the parent company’s 2024 annual report covering all countries where the parent group operates.



Bulgaria

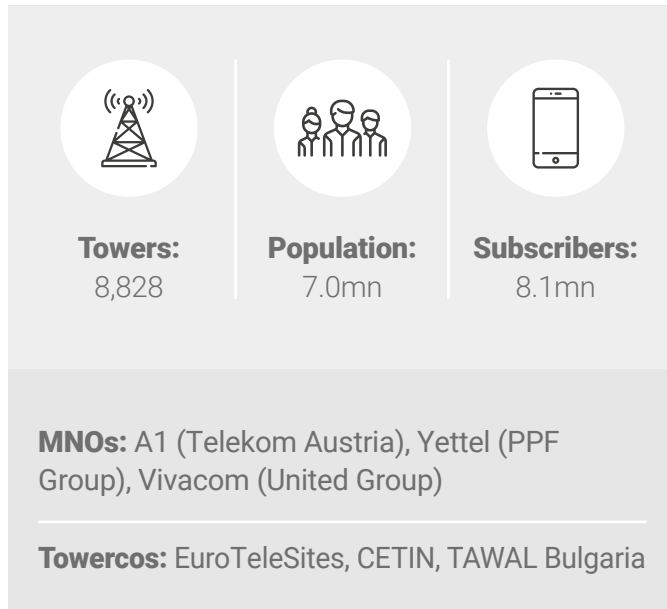
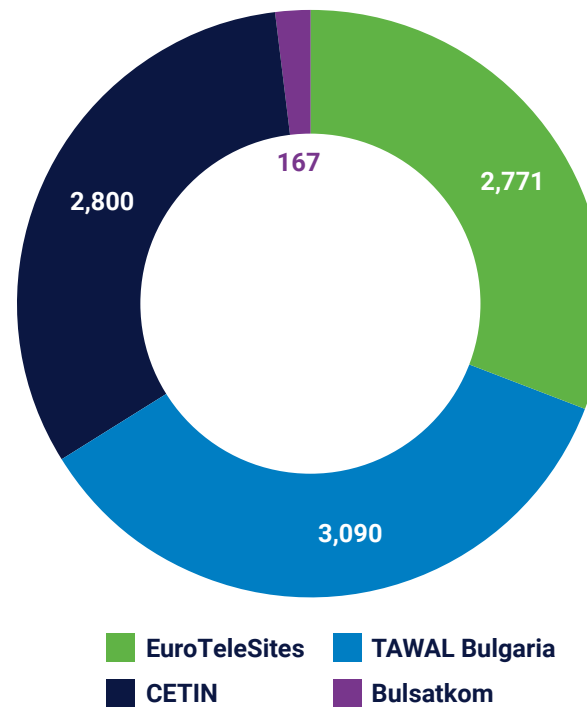


Figure 8: Bulgaria – estimated tower ownership



Source: TowerXchange

was set up following the completion of Telekom Austria Group's tower spin off in Austria in 2023.

In July 2020, Yettel (owned by PPF Group and rebranded

from Telenor Bulgaria) followed in the footsteps of PPF's Czech opco (O2 Czech Republic) and separated out their retail and infrastructure divisions, with their infrastructure carved out into CETIN.

CETIN's remit extends much beyond towers, it takes ownership of the operator's metallic cable pairs, fibre, indoor and outdoor base stations and deploys small cells. In Q4 2021, PPF announced the sale of a 30% stake in CETIN to Singapore's GIC investment fund. In Q3 2023, the PPF Group announced it has agreed a sale of a 50% plus one share in CETIN as well as its MNO units in Bulgaria, Hungary, and Serbia to UAE-based e&. The deal is being investigated by the European Commission.

Vivacom (owned by United Group) is Bulgaria's final operator, and sits behind A1 but ahead of Yettel in terms of market share. In April 2023, United Group announced the sale of its sites in Bulgaria, Slovenia and Croatia to Saudi towerco TAWAL (originally carved out from stc) for EUR1.22bn.

The deal also includes a 20-year Master Service Agreement and additional 2,000 build-to-suit sites. The deal closed in August 2023 and the sites in the three countries are now owned and managed by TAWAL Europe. United Group's MNO subsidiaries in the countries will remain anchor tenants.



Croatia

Towers:
4,336

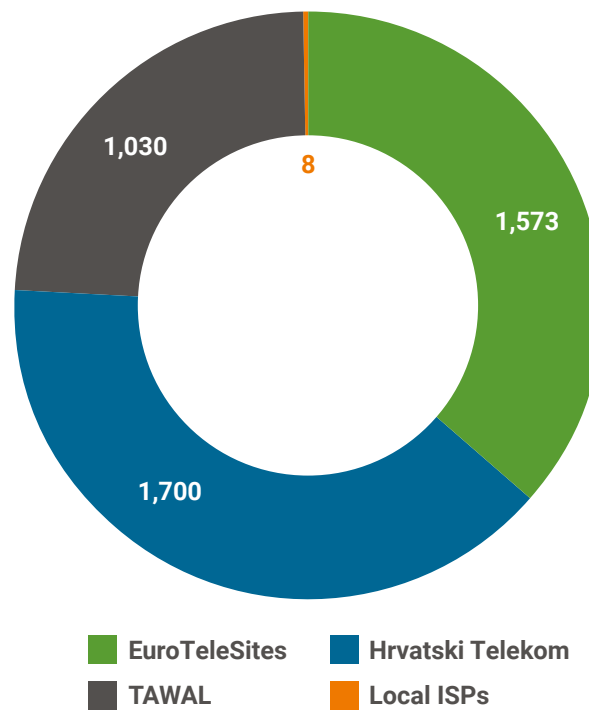
Population:
3.856mn

Subscribers:
4.480mn

MNOs: : A1, Hrvatski Telekom (DT), Telemach (United Group)

Towercos: EuroTeleSites, TAWAL

Figure 9: Croatia – estimated tower count



Source: TowerXchange

There are three MNOs in the country with Hravtski Telekom (HT) having the largest share of 43% and 2.33mn mobile customers. It's a subsidiary of Deutsche Telekom (DT) and in end of February 2025 announced it is starting the process to carve-out its passive infrastructure into a wholly owned subsidiary. The process is expected to complete in Q3 2025.

This is the second DT subsidiary in CEE examining tower carve out options, following Greek OTE's announcement earlier in 2024.

A1 is the second largest (mobile and fixed line) operator with approximately 2mn subscribers. Its parent company A1 Telekom Austria Group has completed the carve out

of its Croatian towers. The assets, alongside sites in Austria, Bulgaria, Serbia, Slovenia and North Macedonia are managed by the EuroTeleSites towerco which is headquartered in Vienna and was listed on the Vienna Stock Exchange in September 2023.

A1 is the anchor tenant but the towerco is looking to expand its customer base. Following the carve-out and towerco set up, an additional 118 sites from A1 Group have been sold to EuroTeleSites. Of these, 25 are in Croatia, 85 are in North Macedonia and eight in Slovenia. EuroTeleSites reports 53% of sites are macros and 47% rooftops and a 1.15x tenancy ratio in the country.

Telemach (acquired by United Group in 2019) is the third MNO with the smallest market share. In August 2023, United Group completed the sale of its sites in Bulgaria, Slovenia and Croatia, including the Telemach towers, to Saudi towerco TAWAL (originally carved out from stc) for EUR1.22bn.

In addition to the three MNOs, MVNOs Tomato (owned by A1) and Bonbon (owned by HT and running on its network) provide mobile services. Following the 5G frequency auctions that completed in 2023, network densification in rural and suburban areas will be taking place, including small cell deployment in urban areas.



Cyprus

Towers:
1,008

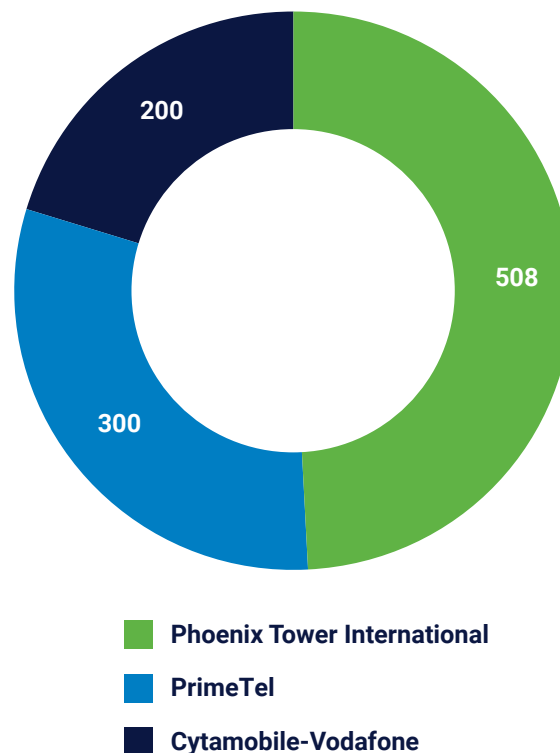
Population:
0.8mn

Subscribers:
1.3mn

MNOs: Cytamobile-Vodafone, Epic (Monaco Telecom), PrimeTel (Signal Capital Partners)

Towercos: Phoenix Tower International

Figure 10: Cyprus – estimated tower count



Source: TowerXchange

In mid-2022, Cytamobile-Vodafone announced the completion of the deployment phase of its 5G mobile network claiming Cyprus to be the first country in the

EU to have reached 100% 5G population coverage. Epic is the second largest operator with a 38.8% market share. Formerly MTN Cyprus, it rebranded to Epic after it was acquired by Monaco Telecom in mid-2018 for EUR260mn (USD\$284mn). Monaco Telecom is owned by the Principality of Monaco and French billionaire Xavier Niel's firm NJJ Holdings.

In March 2021, Phoenix Tower International announced its acquisition of Epic's passive infrastructure in the country which was part of a sale and leaseback deal covering the countries of Malta and Cyprus.

It included a 20-year leaseback agreement for the MNO as well as a build-to-suit programme for an undisclosed number of sites.

PrimeTel which launched as the third MNO in 2015 following several years of operating as an MVNO, has the smallest market share, standing at approximately 11%.

It was placed under administration in July 2023, after it defaulted on a loan it secured from London-based private equity fund Signal Capital Partners. Cablenet, a subsidiary of Maltese MNO GO, operates as the only MVNO in the country using the network of Cytamobile-Vodafone.

Cypriot MNOs operating in the southern part of the island are Cytamobile-Vodafone, Epic and PrimeTel.

Cytamobile-Vodafone is the longest-established operator with approximately a 49.8% market share by subscribers. It's fully owned by the state and offers mobile services under the Cytamobile-Vodafone brand since 2004 following a partnership agreement with Vodafone Group. The government is understood to be pressing for a sale of a minority stake.



Czech Republic

Towers:
13,868

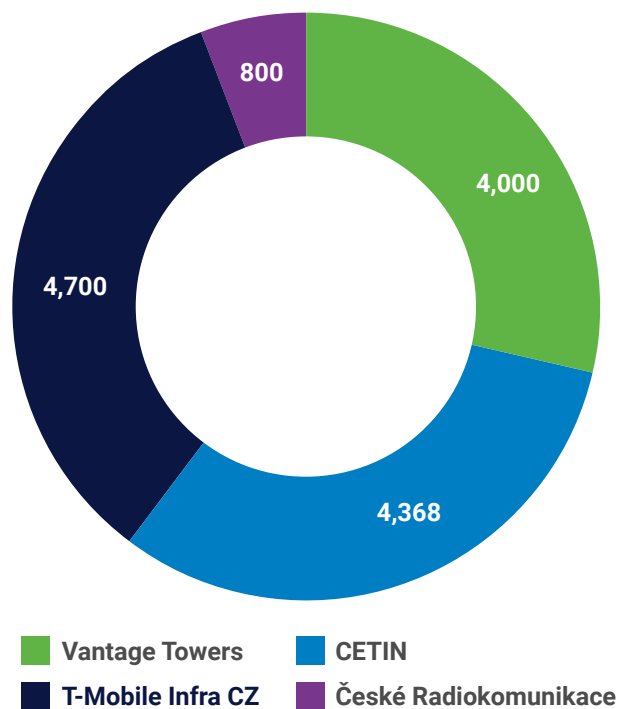
Population:
10.7mn

Subscribers:
13.2mn

MNOs: O2 (PPF Group), T-Mobile (Deutsche Telekom), Vodafone

Towercos: České Radiokomunikace, CETIN, T-Mobile Infra CZ, Vantage Towers

Figure 11: Czech Republic – estimated tower count



Source: TowerXchange

O2 is the Czech Republic's leading MNO. When PPF acquired O2 Czech Republic from Telefónica in 2014, they immediately set about separating out the retail assets from the infrastructure, in the latter case creating CETIN. CETIN's remit extends beyond towers, with CETIN owning O2's metallic cable pairs, fibre and indoor and outdoor base stations. PPF has since separated out its infrastructure in Bulgaria, Hungary and Serbia combining the assets with those in the Czech Republic to form CETIN Group and in Slovakia into O2 Networks. In Q3 it announced the sale of

a 50% plus one share of its MNO and infrastructure units only in Bulgaria, Hungary, Serbia and Slovakia to UAE-based e&. Singapore's GIC investment fund owns a 30% stake in CETIN. CETIN and T-Mobile CZ have a network

sharing agreement including 2,300 sites and parts of the active radio equipment with CETIN owning and operating the network in the eastern part of the country, and T-Mobile serving the western part. The agreement excludes the two major cities – Prague and Brno. Vodafone has carved out their tower portfolio into their pan-European towerco, Vantage Towers. Vantage Towers became operational in summer 2020 and has a portfolio of 4,000 sites in Czech Republic.

T-Mobile CZ, owned by Deutsche Telekom, has carved out its passive infrastructure assets in the country (towers and rooftops) which are managed by T-Mobile Infra CZ. At group level, Deutsche Telekom has formed a towerco business unit – GD Towers – under which its German towerco, Deutsche Funkturm, and its Austrian towerco, TOWERS Infra Austria, are held. In the Netherlands, Deutsche Telekom's opco, T-Mobile has merged their towers with Cellnex Netherlands, forming a new infra fund, DIV, which holds a 38% stake in Cellnex Netherlands as a seed asset. DIV plans to invest in further digital infrastructure portfolios.

In addition to Vantage Towers, CETIN and T-Mobile, broadcast towerco, České Radiokomunikace (acquired by Cordiant Digital Infrastructure) owns around 800 towers in the Czech market and leases some of them to CETIN and T-Mobile Infra CZ.

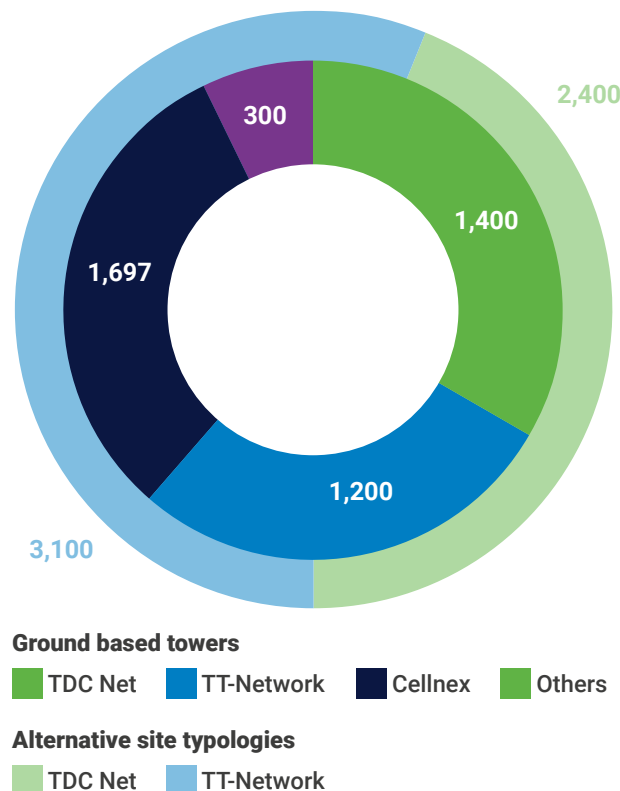
Denmark



There are four MNOs in the Danish market. Market leaders, TDC, have separated out their service and infrastructure divisions, with TDC Net now managing the operator's infrastructure for fixed and mobile networks (and Nuuday managing the service side). TDC Net have around 1,400 towers and a further c. 2,400 sites of other typologies (rooftops, chimneys, windmills etc). It is understood that rooftop and other such sites are generally not shareable in the Danish market, with each operator having to sign their own agreement with the landlord and as such, those 2,400 sites are single tenanted.

3, owned by Hutchison, have sold their towers to Cellnex as part of a [pan-European deal](#). Cellnex has around 1,600

Figure 12: Denmark – estimated tower count



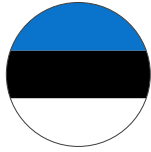
Source: TowerXchange

towers in the market, Cellnex has a build to suit pipeline of 500 further sites. Telenor and Telia (now owned by energy-telco group Norlys) share their networks and spectrum

through the 50:50 joint venture infraco, TT-Networks which was established back in 2012. Telia has completed the sale of its Denmark business to Norlys for an enterprise value of DKK 6.25bn (USD\$882.48mn) including the infraco. It has overhauled its board following the takeover with Ulf Lund, Director of Stakeholder Relations at Norlys becoming the new chairman of TT Network. TT-Networ is also involved in IBS and provides back-up power on towers.

Whilst Telia has carved out its towers in Sweden, Norway and Finland into towerco business units held under a new entity, Telia Asset Management (having sold a 49% stake in each to Brookfield and Alecta), TT-Networ was excluded from Telia Asset Management's perimeter. [Telenor has similarly created towerco business units](#) across its Nordic operations, grouping these together under Telenor Tower Holdings.

Whilst TT-Networ was initially included in Telenor Tower Holdings' perimeter, it is not included in financial reporting around the tower business. With TT-Networ's remit having been extended to 5G, there is no indication that these developments will impact the future of the joint venture. "The future, as part of Norlys, has not yet been mapped out" Telia has said. In addition to the sites owned by Cellnex, TT-Networ and TDC Net, there are around 300 further towers in the Danish market.



Estonia

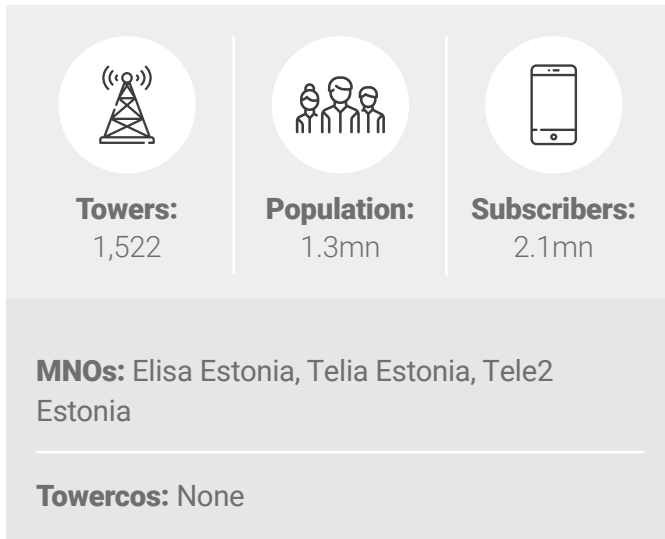
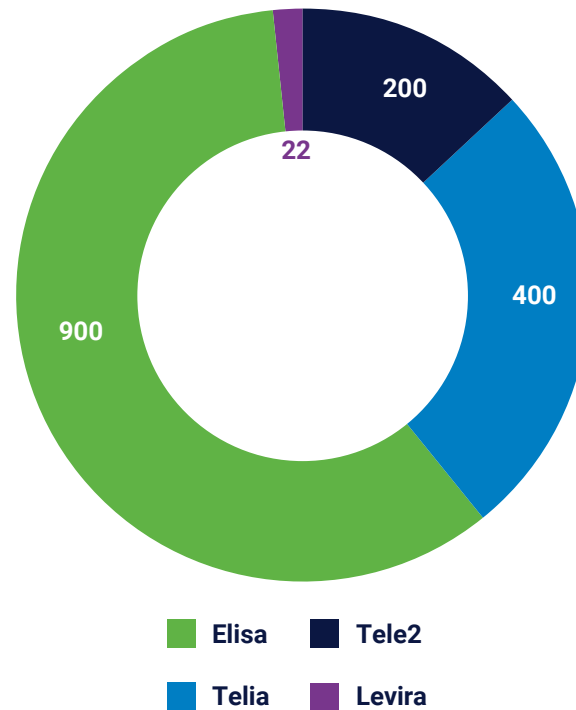


Figure 13: Estonia – estimated tower count



Source: TowerXchange

Telia Estonia is the second MNO with approximately 35% share. Its fixed broadband network, a mix of fibre and copper, reaches over 550,000 households. It's

announced further plans to work on fibre development projects in 2023 in big cities, including the capital. Tele2 has an approximately 23% market share. It also provides broadband services over the fibre network of power supplier Elektrilevi. The operator is rumoured to be working with Perella Weinberg Partners to explore divestment of its towers in Estonia, Latvia and Lithuania, followed by a sale.

All three 5G spectrum pioneer bands have been assigned with the most recent one being in the 26GHz band in May 2023. The country's MNOs have confirmed they're working on expansion of their 5G infrastructure.

A growing trend related to reaching certain ESG goals among Estonian telecoms providers is the use of renewable energy. Elisa and Telia have started a long-term partnership with local renewable energy solutions developer, Sunly City to install solar panels next to their towers to increase green energy consumption.

Towers in the country are shared without any legal requirement or a clear pattern - sharing happens where possible and reasonable.

TowerXchange understands there is increasing interest in the telecommunications assets in Estonia where at present there are no towercos.

There are three MNOs – Elisa, Telia and Tele2. Elisa Estonia has the highest market share with 42% of subscribers. Elisa Estonia's parent company Elisa Group acquired the Finnish and Estonian units of IT security, data centre and network integration and design services provider Santa Monica Networks for an undisclosed amount in 2017. Elisa Estonia now offers IT services to B2B customers such as security and teleworking solutions.



Finland



Towers:
9,805



Population:
5,541,020



Subscribers:
7,150,000

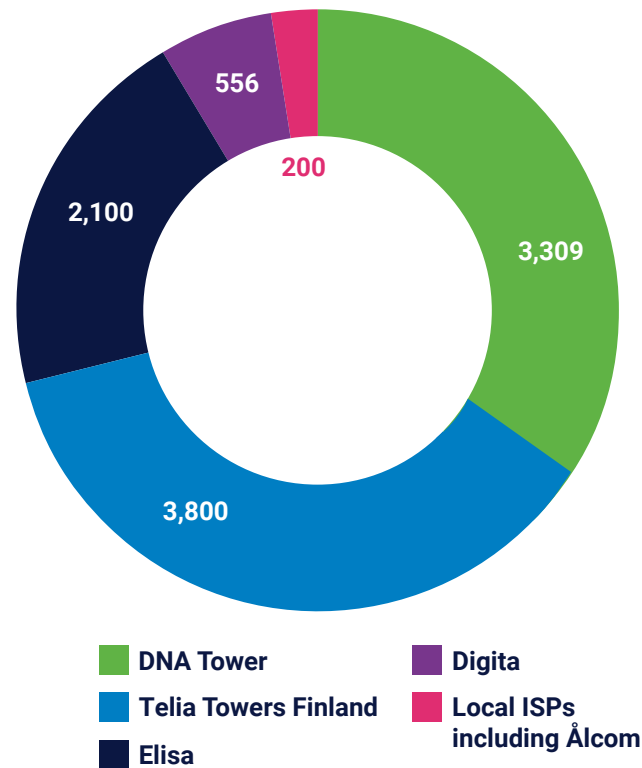
MNOs: Elisa, Telia Finland, DNA (Telenor)

Towercos: Digita, Telia Towers Finland, DNA Tower

There are three MNOs in one of the most sparsely populated countries in Europe - DNA, Elisa and Telia Finland. Elisa has the highest market share, sitting at around 39%. Telia Finland is the second MNO in the market with approximately 31% share, but it has seen its share decline from 34% as DNA adds customers. In addition to expanding 5G coverage, Telia and Elisa are working with Eltel on the construction of FTTH infrastructure.

DNA, which is wholly owned by Telenor, is the country's third mobile operator with approximately 30% market share. Telenor acquired a majority stake of 54% in the MNO in April 2019 for €1.9bn and later in the year increased its share to 94.3%.

Figure 14: Finland – estimated tower count



Source: TowerXchange

DNA and Telia have a JV called Suomen Yhteisverkko (Finnish shared network) which manages, designs and maintains the mobile network in Northern and Eastern Finland. DNA owns

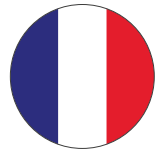
49% in Suomen Yhteisverkko and Telia Finland owns 51%, with both sharing spectrum as well as sites.

Digita is Finland's "oldest" towerco which also owns and operates a digital terrestrial television and radio broadcasting tower network. Founded in 1999 as a spin-off from national broadcaster Yle, it was acquired by DigitalBridge in 2019 alongside 556 towers for an undisclosed amount.

Telia Towers Finland was carved out just after Sweden-headquartered Telia announced it was exploring the potential monetisation of its tower portfolio back in their Q4 2020 results. Telia Towers Finland manages only macro towers with single tenanted rooftop sites and stakes in joint ventures remaining outside of the towerco's perimeter.

DNA Tower, the youngest towerco in Finland, was formed in 2022 and is wholly owned by DNA.

Finland is the first country in Europe where the concept of using telecom infrastructure as a virtual power plant - supplying renewable energy generated onsite, storing it and sending it back to the country's electricity grid as well as grid balancing - is implemented. Initially developed and operated by Elisa, it is now being adopted by DNA Tower on some of its sites.



France



Towers:
62,547



Population:
67.1mn



Subscribers:
72.0mn

MNOs: Bouygues Telecom, Free Mobile (Iliad), Orange, SFR

Towercos: ATC Europe, Cellnex, TOTEM, Phoenix Tower International, TDF, Towercast

Three of France's four MNOs have completed tower sales in the country, whilst Orange has carved out their portfolio into their own towerco, TOTEM. TOTEM's perimeter is initially confined to France and Spain (with a view to explore the integration of further assets at a later date). TOTEM currently has 19,859 towers in France, of which 58% are ground based towers, 30% flat rooftops and 12% other locations.

Bouygues Telecom was one of Europe's first MNOs to sell towers to an independent towerco, selling 2,166 of their

estimated 17,000 towers to Antin Infrastructure Partners' FPS Towers in 2012 for €185mn (originally retaining 15% equity, which Antin subsequently bought in 2015). FPS Towers expanded their portfolio to over 2,400 sites and secured the rights to 20,000 rooftops and 76,000 electricity pylons.

In 2016, Antin subsequently [sold FPS Towers to ATC Europe](#) (American Tower's European arm) for a total of €697mn. As of Q2 2023, ATC Europe has a portfolio of 4,281 sites in the country. A contract with Orange to rollout up to 2,000 new sites, primarily in rural areas and along transport routes, gives ATC Europe strong growth potential in the market.

Bouygues Telecom has also completed several tower transactions with Cellnex. In 2016 they sold 500 towers ([an initial tranche of 230 towers](#), followed by a second tranche of 270) to Cellnex for a total of €147mn; in February 2017 they [signed a third deal](#) with the aforementioned governing the transfer of 1,800 urban (primarily rooftop) sites and 1,200 new build towers for a total of €854mn; later the same year they signed a further deal with Cellnex transferring a further 600 sites for €170mn.

In December 2018, the operator deepened their relationship with Cellnex, signing an agreement for the

towerco to provide distributed processing capabilities (edge computing) at 88 strategic telecom centres, a deal since extended to cover 150 centres. In February 2020, [Cellnex and Bouygues' relationship deepened even further](#) with the two parties announcing the formation of a joint venture to deploy a nationwide fibre network, with plans to invest up to €1bn to rollout up to 30,000 km of fibre, with the new agreement also including the rollout of a further 90 strategic telecom centres.

In March 2020, Bouygues Telecom announced a second joint venture, this time with [Phoenix Tower International \(PTI\)](#), with the PTI controlled joint venture set to rollout up to 4,000 new sites outside Very Dense Areas as part of Bouygues' New Deal Mobile obligations.

In 2018, MNO SFR (owned by Altice) [carved out their towers into a new towerco](#), SFR Towerco, in which they subsequently sold a 49.99% stake to investment firm, KKR and rebranded the unit to "Hivory". In February 2021, Cellnex announced that it had reached a deal to acquire Hivory and its 10,500 sites, with plans to deploy a further 2,500 sites over the next eight years.

The deal closed in Q4 2021, with one of the conditions of the French Competition Authorities approval being that Cellnex must divest 3,200 sites. Cellnex has reached a deal



FRANCE

to sell the towers to Phoenix Tower. The asset transfer is currently underway with 2,353 sites now sold to PTI for EUR 631mn and is expected to complete in 2024.

Altice is expected to explore a stake sale in SFR to reduce its EUR23 bn debt of its French unit. The move will be in parallel to the sale of its datacentre unit.

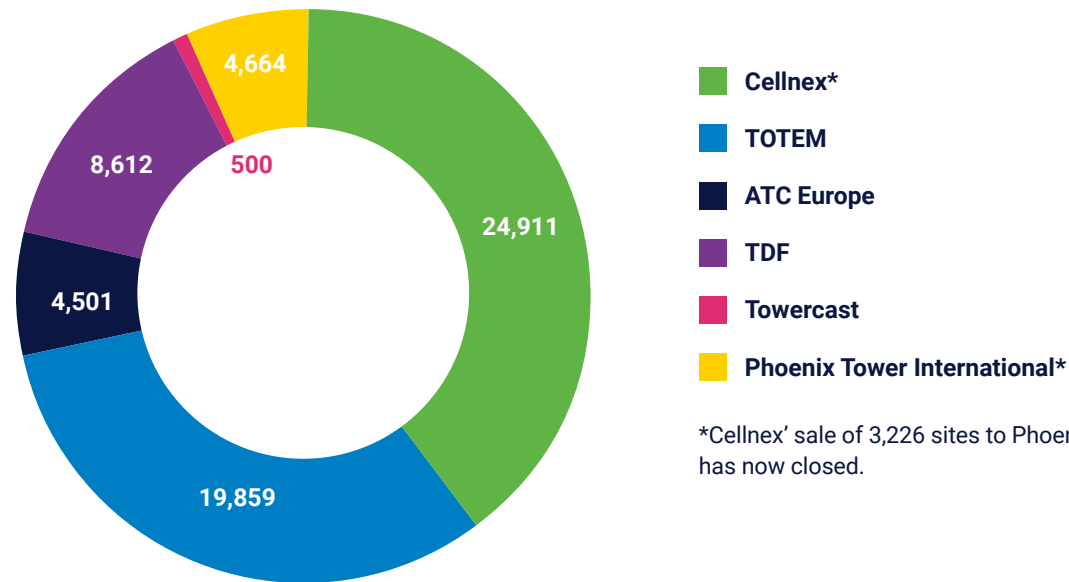
In mid-2019 France's newest MNO, Free Mobile (owned by Iliad) also agreed a tower sale in the country, [reaching a deal with Cellnex](#) for the sale of its 5,700 sites, retaining a 30% stake in the assets (Cellnex has since exercised the right to acquire the Iliad's 30% stake in the assets). The transaction also involves a build to suit component with Cellnex to add up to 2,500 new sites for the operator.

Broadcast-telecom hybrid TDF has slightly over 8,600 telecom towers as well as an established broadcast business and growing fibre deployment. In 2016, TDF acquired the previously family owned, towerco, ITAS TIM and its 420 towers for a reported €100mn.

Brookfield, which owns a 45% stake in TDF has appointed advisors to oversee the sale of its shareholding – the investor having previously attempted a sale in 2019 and 2021.

Broadcast towerco, TowerCast (owned by NRJ Group) which owns around 500 sites also sell co-locations to telecom clients in the country. NRJ announced that

Figure 15: France – estimated tower count



Source: TowerXchange

TowerCast would come to market in 2017, but the deal has quietly slipped off the radar since.

Some reports suggest that France may need as many as 50,000 new points of presence, with the regulator, ARCEP, stipulating that the three most established operators (Orange, SFR and Bouygues) must add a further 15,000 points of presence between them.

The newest market entrant, Free Mobile, needs to add a further 10,000 points of presence to compete with its

peers. Co-location on existing structures will help hit the PoP targets but new build is still expected. Cellnex is committed to add up to 5,020 sites through its BTS agreements with Bouygues, Free and SFR. PTI plans to build up to 4,000 new sites for Bouygues in rural areas and ATC Europe building rural sites for Orange. Free Mobile signed a long-term agreement with TOTEM in Q4 2022 which gives the MNO access to TOTEM's ground-based and roof top sites in France. The French government is also heavily incentivising MNOs to build towers in rural areas, which should further promote BTS.



Germany



Towers:
79,746



Population:
83.1mn



Subscribers:
107.2mn

MNOs: O2 (Telefónica), Telekom (Deutsche Telekom), Vodafone, 1&1

Towercos: ATC Europe, Deutsche Funkturm, Media Broadcast Group, Phoenix Tower International, Vantage Towers, 5G Synergiewerk

Market leading mobile network operator, Telekom (owned by Deutsche Telekom) founded their infraco Deutsche Funkturm in 2002, initially as a subsidiary but then restructured into a “sister company” in 2017 with a view to commercialise their assets more actively. Deutsche Funkturm (DFMG) is held under Deutsche Telekom’s GD Towers unit, alongside Austrian towerco, TOWERS Infra Austria, with DFMG helping to steer the new entity. In July 2022, Deutsche Telekom reached a deal to sell a 51% stake in GD Towers to DigitalBridge and Brookfield which closed

in Q1 2023. Deutsche Telekom retains a 49% stake in the entity along with significant majority protection rights, including the ability to appoint two of GD Towers’ five shareholder committee members, including the Chairman. GD Towers has contracts in place to rollout 5,400 new sites for DT’s opcos in Germany and Austria, with the bulk of such rollout set to take place in Germany.

In 2016, [Telefónica carved their 2,350 German ground based towers into a newly created infraco, Telxius](#), in a deal valued at €587mn (with the operator also transferring Spanish and CALA towers as well as 31,000km of subsea cabling into the infraco). Telefónica subsequently sold a 40% stake in Telxius to investment firm KKR, before selling an effective 9.99% stake to Pontegadea, leaving the operator with a controlling 50.01% stake in the infraco. In June 2020, [Telefónica announced the sale of a further 10,100 \(primarily rooftop\) sites to Telxius](#) for €1.5bn, with the deal also including a BTS commitment for Telxius to add up to 2,400 (primarily greenfield) new sites for the operator. In January 2021, American Tower announced that it had reached a deal to acquire Telxius’ tower business. The deal has since closed adding a further 12,500 sites to American Tower’s German portfolio as well as a build to suit pipeline of 2,400 sites.

American Tower entered the German market back in

2012 with the acquisition of KPN’s 2,000 E-Plus towers for €393mn (KPN having since sold E-Plus to Telefónica). American Tower’s German sites were transferred to ATC Europe, a joint venture formed with PGGM in which the towerco had a controlling 51% stake. Following the Telxius deal, American Tower sold a 48% non-controlling interest in the newly enlarged ATC Europe to CDPQ (30%) and Allianz (18%), with PGGM converting its prior holding in ATC Europe to minority stakes in the towerco’s local German and Spanish operating companies.

Vodafone sold an 81.7% of its stake in its European toweco Vantage Towers to a consortium of investment firms KKR & Co., Global Infrastructure Partners (GIP) and The Public Investment Fund of Saudi Arabia (PIF). The co-control partnership of the MNO, GIP and KKR called Oak Holdings has bought out the minority shareholders in Vantage Towers and now holds a 89.3% stake. The towerco was delisted at the end of 2023.

In late 2021 Vantage signed an agreement with new MNO market entrant 1&1 to provide at least 3,800 (but up to 5,000) existing sites to 1&1 throughout the country for the next 20 years. 1&1 missed coverage targets claiming Vantage Towers has failed to activate 1&1 sites on its infrastructure, leaving the MNO with just five operational 5G sites at end-2022, behind the 1,000 it was required to

GERMANY

have under the terms of its spectrum licences. In August 2023 it announced a national roaming agreement with Vodafone's 5G network in areas that are not yet covered by its own network starting from July 2024 ousting current network partner Telefonica and gradually reducing services. In December 2023 1&1 has announced the activation of its own mobile network and the start to offer services to customers on it.

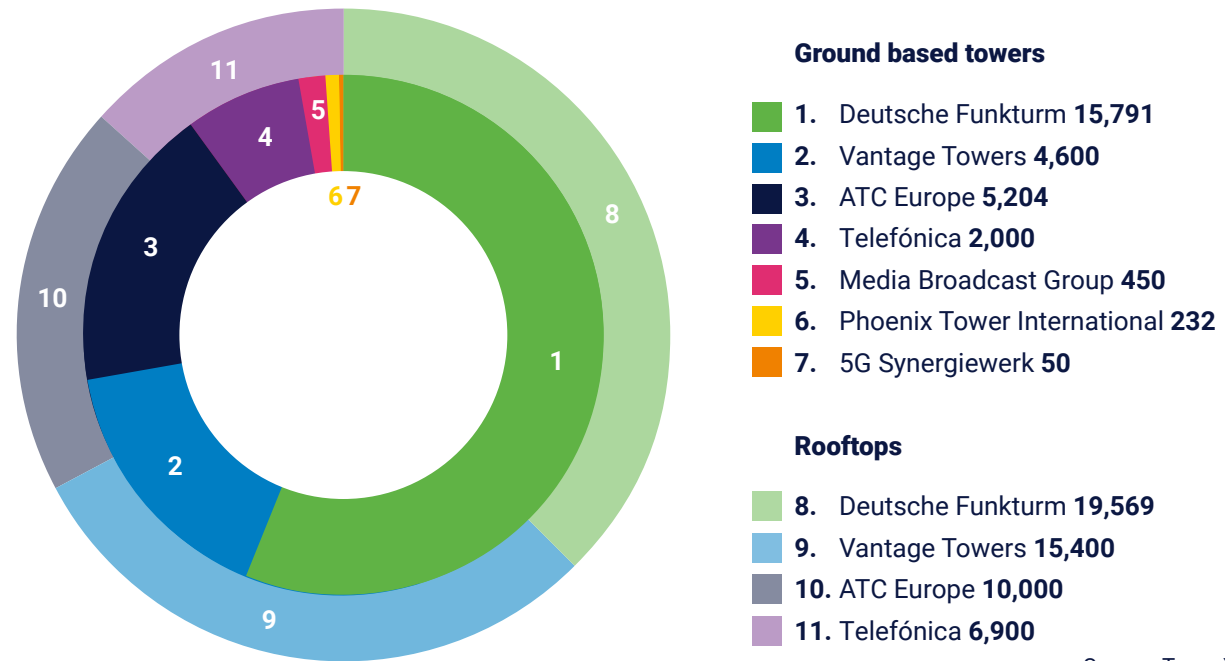
German broadcast towerco Media Broadcast Group owns a further 450 towers in Germany, and was acquired by Freenet in 2016 for €295mn (around 12x EV/EBITDA).

In October 2023 Phoenix Tower International entered the market announcing it's buying 220 sites from Novec GmbH, a subsidiary of Netherlands-based Novec BV.

Whilst co-location potential remains strong on Germany's ground-based towers (DFMG reports a 2.3x co-location ratio, more recently formed Vantage Towers 1.8x), unique conditions in the German market means that the tenancy ratio on rooftop sites sits at almost exactly one. Landlords in Germany typically agree rooftop leases on a per operator basis rather than per site basis and as such, the synergies available from passive sharing on rooftops are greatly reduced. Towercos are working on lease renegotiations to improve the economics.

In November 2019, Deutsche Telekom, Telefónica and Vodafone announced an initiative to jointly cover white spots (primarily in rural areas and along transport

Figure 16: Germany – estimated tower ownership



Source: TowerXchange

routes) which would otherwise be uneconomic to cover. Each operator will build 2,000 sites which will be made available for the other two players to use via passive sharing agreements.

Vodafone, Deutsche Telekom and Telefónica have also teamed up to improve LTE, 4G and 5G coverage in rural areas and along traffic routes at so called "grey spots" – areas where LTE quality coverage is only available from a single operator. The operators will implement active

network sharing across antenna sites using a MOCN approach. Active network sharing agreements are unlikely to be expanded beyond this. New build in the German market remains significant. Vantage Towers has a commitment in place with Vodafone to add 5,500 towers (2,000 of which under the white spot programme) and reports demand outstripping this, Deutsche Telekom rolled out 1,400 new sites in 2021 and over the past four years have built 6,300 new sites. American Tower is set to add up to 2,400 new sites for Telefónica.



Greece

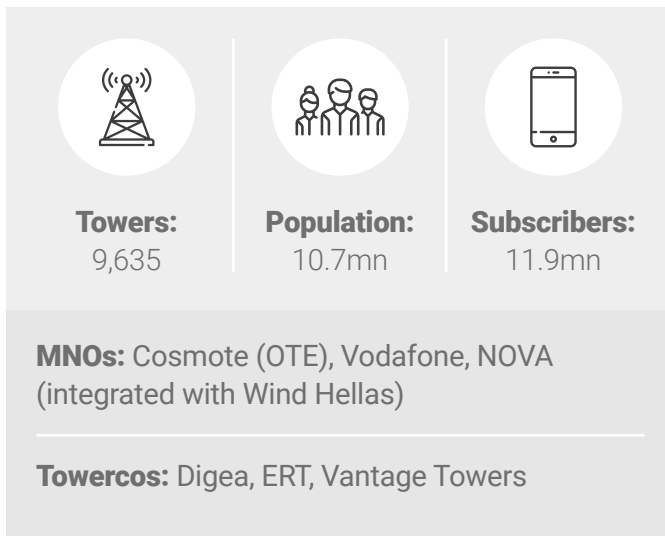
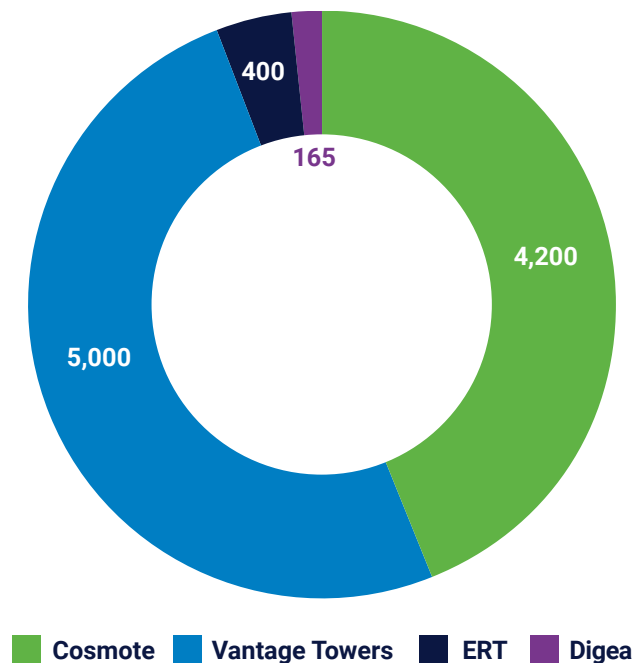


Figure 17: Greece – estimated tower ownership



Source: TowerXchange

In July 2019, Vodafone announced plans to carve out its European towers into a towerco, Vantage Towers. In Greece, Vodafone shared passive infrastructure with Wind Hellas via the joint venture, VICTUS Networks (although the towers remained on the balance sheets of the MNOs).

In July 2020 in an announcement surrounding the formal launch of Vantage Towers, Vodafone announced that it had reached a deal with Crystal Almond “(the former controlling shareholder of Wind Hellas) for the two operators to transfer their tower assets to Vantage Towers Greece. Whilst initially a joint venture between the two operators (with Vodafone having the majority share), the IPO of Vantage Towers triggered the automatic acquisition of Crystal Almond’s shareholding by Vodafone, with Vantage Towers Greece now wholly owned by Vodafone’s Vantage Towers.

Vodafone’s sale of its 81.7% stake in Vantage Towers to a consortium of investment firms KKR & Co., Global Infrastructure Partners (GIP) and The Public Investment Fund of Saudi Arabia (PIF) completed in Q1 2023 with the new co-control partnership of the MNO and investment firms called Oak Holdings. Vantage Towers is the preferred supplier of towers to both Vodafone Greece and NOVA (former Wind Hellas), with the two parties having committed to being anchor tenant on 250 new sites (500 in total) to be built by Vantage Towers in the next five years. In addition to sharing passive infrastructure through Vantage Towers, Vodafone and NOVA (former Wind Hellas) have an active sharing agreement in place. Beyond Cosmote and Vantage Towers Greece, broadcast towerco, Digea owns 165 towers in the Greek market, with national broadcaster ERT owning 400 sites.

Hungary

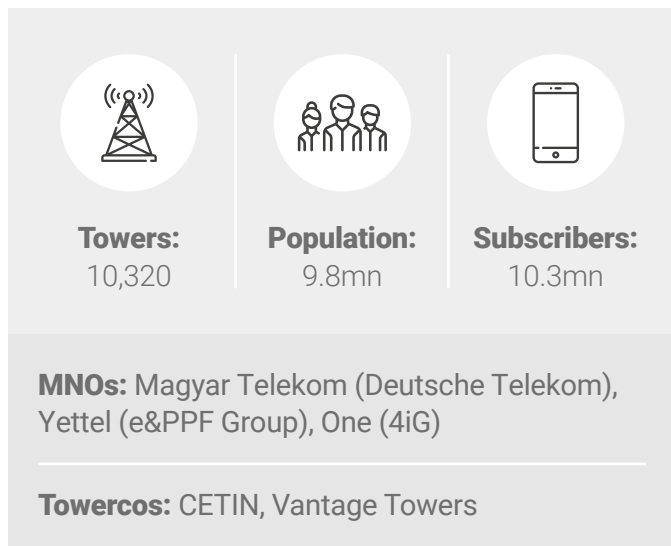
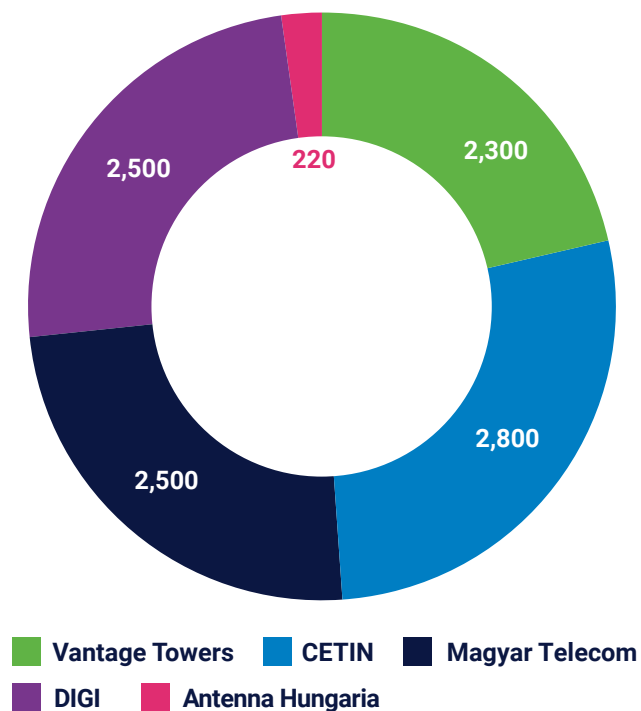


Figure 18: Hungary – estimated tower ownership



Source: TowerXchange

with its strategy for infrastructure separation. The deal is expected to complete in 2024.

Vodafone has carved out their tower portfolio into their pan-European towerco, Vantage Towers which has a portfolio

of 2,100 sites in Hungary. In July 2020, Telenor Hungary (now Yettel) followed in the footsteps of PPF's Czech opco (O2 Czech Republic) and separated out their retail and infrastructure divisions, with their infrastructure carved out into CETIN. CETIN's remit extends much beyond towers, with CETIN taking ownership of the operator's metallic cable pairs, fibre and indoor and outdoor base stations. In Q4 2021, PPF agreed the sale of a 30% stake in CETIN to Singapore's GIC investment fund.

Magyar Telekom, the Hungarian incumbent, is 59.3% owned by Deutsche Telekom which has not yet announced carve out plans for its assets in the country. Romanian owned DIGI launched mobile operations in Hungary in 2019, although the operator was excluded from the country's 5G auctions. It sold its Hungarian business to 4iG and exited the market in early 2022. Broadcast towerco, Antenna Hungaria (which is also owned by 4iG) is understood to own around 220 towers in the country and also leases space to telecom operators. It owned a 25% stake in both Yettel and CETIN but in Q1 2023 Antenna Hungaria swapped its 25% stake in the Hungarian subsidiaries of PPF Telecom Group for 19.5% of Vodafone shares, increasing its stake in Vodafone Hungary to 70.5% and ceasing to have any shareholding in Hungarian Yettel and CETIN.



Iceland

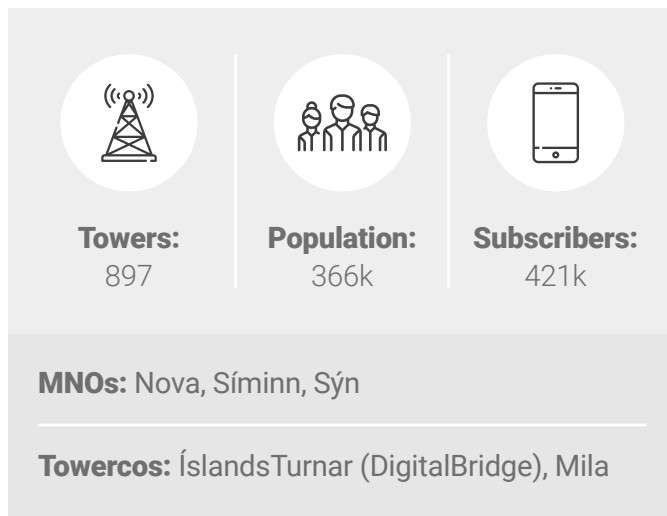
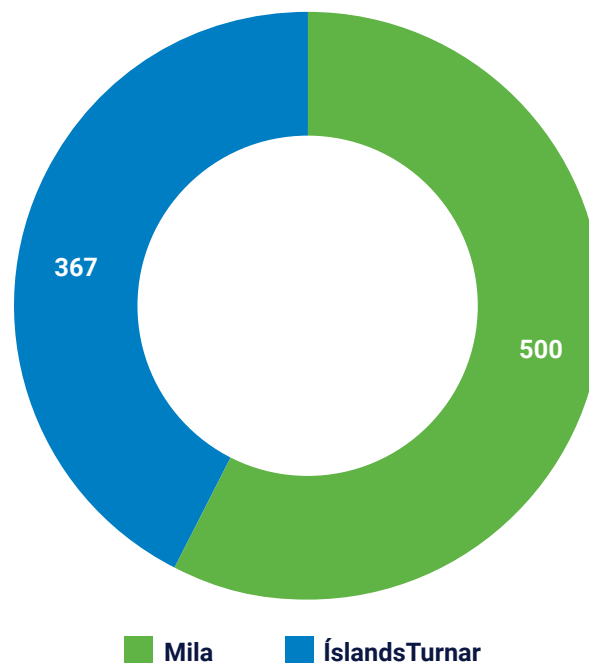


Figure 19: Iceland – estimated tower ownership



Source: TowerXchange

There are three major operators in the Icelandic market - Síminn, the incumbent; Sýn and Nova (which was the newer challenger in the market). Whilst Síminn marginally has the largest market share, the subscriber base is relatively evenly split between the three operators.

In 2007, Síminn carved out its infrastructure assets into a dedicated infraco, Míla. Míla owns a comprehensive network comprised of fixed copper and fibre access, backbone and connectivity, as well as active equipment across a nationwide footprint. Their tower count is understood to be around 500 sites. In 2021, investment firm

Ardian, reached a deal to acquire 100% of Míla, although the deal is yet to receive regulatory approval with the Icelandic Competition Authority having voiced concerns. In late 2021, Sýn and Nova simultaneously announced the sale of their tower portfolios to ÍslandsTurnar, a new

towerco formed by DigitalBridge. The deal has since closed, making ÍslandsTurnar Iceland's first independent towerco. ÍslandsTurnar's site count stands at 367 sites and includes both ground based and rooftop sites. The deal also included a build to suit component for an undisclosed number of sites over a long-term period. Recently, Sýn agreed the sale of its core network to open access infrastructure provider, Ljosleidarinn.

Like much of the Nordics, active sharing is a feature of the Icelandic market with Sýn and Nova have an extensive national RANsharing agreement in place. As such, nearly all of ÍslandsTurnar's 367 sites have both operators present on them. Whilst there are some tenancies from Síminn on the ÍslandsTurnar portfolio, Síminn has historically preferred to use its own infrastructure when rolling out its network.

As the incumbent, regulation has been applied to Síminn/ Míla requiring them to allow other operators access to their tower portfolio at a reasonable rate. As such, Míla has tenancies from both Sýn and Nova on their towers.

5G rollout is in the early stages in the country and whilst it is early days, there are very clear rollout and investment plans in place from the country's operators providing significant growth opportunities for both ÍslandsTurnar and the soon to be independent, Míla.



Ireland



Towers:
6,004



Population:
4.9mn

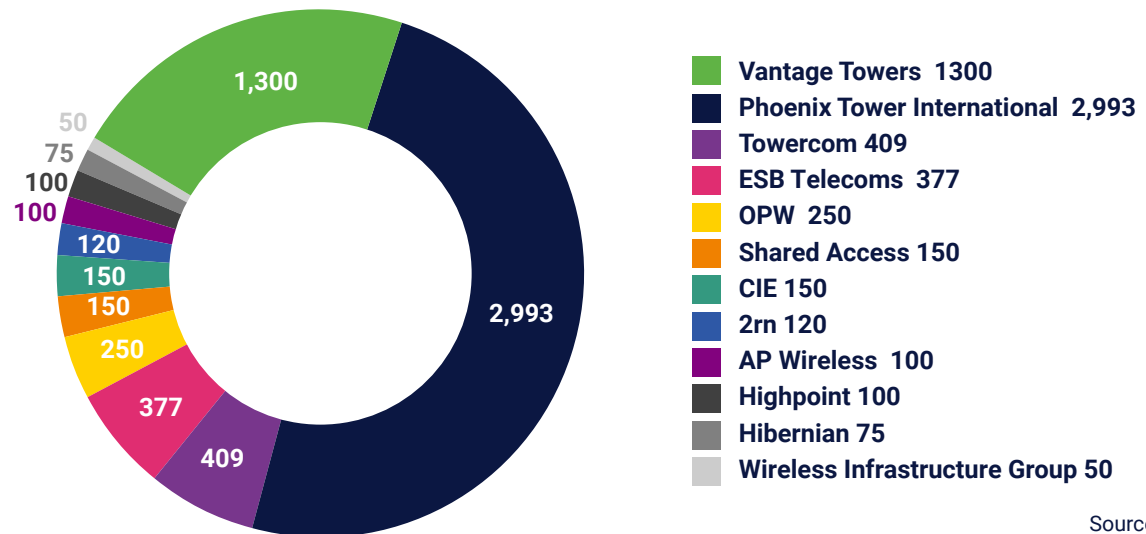


Subscribers:
5.1mn

MNOs: 3 (CK Hutchison), eir, Vodafone

Towercos: 2rn, Hibernian, Highpoint (Obelisk), Phoenix Tower International, Shared Access, Towercom, Wireless Infrastructure Group, Vantage Towers

Figure 20: Ireland – estimated tower ownership



Source: TowerXchange

There are three MNOs in the Irish market. Vodafone, 3 and eir. Vodafone carved out their towers into Vantage Towers, and in Q1 2023 completed the sale of its 81.7% stake to investment firms KKR and GIP, forming a co-control partnership with them called Oak Holdings. In May 2020, [eir signed a deal](#) to sell its portfolio of 650 sites to Phoenix Tower International for €300mn. The deal also included a BTS commitment with PTI expected to add up to 800 new sites for the operator. In early 2024 PTI acquired 100% of Cellnex' business in the country for EUR971mn. The deal has now closed. Three (owned by

CK Hutchison) has sold its towers to Cellnex as part of a [pan-European deal](#). Cellnex entered the Irish market back in 2019 with the [acquisition of Cignal](#) (Cignal having previously acquired the towers – and the land under third party towers – of state forestry company, Coillte back in 2015).

Its recent Irish market exit following the sale of all assets to PTI follows a strategy to deleverage. There have been a handful of other tower transactions in the Irish market, most notably the acquisition of 340 masts from Eircom (owners

of eir) by Towercom back in 2007. In July 2023 investment firm John Laing acquired Towercom for an undisclosed amount. Over half of Towercom's 409 sites are to be upgraded or redeveloped over the next 10 years.

There is a long tail of other independent towercos in the Irish market (notably Brookfield backed Wireless Infrastructure Group which, whilst just having a portfolio of 50 sites in Ireland, has a portfolio of 3,200 towers in the UK).

Italy



Towers:
54,882



Population:
60.3mn

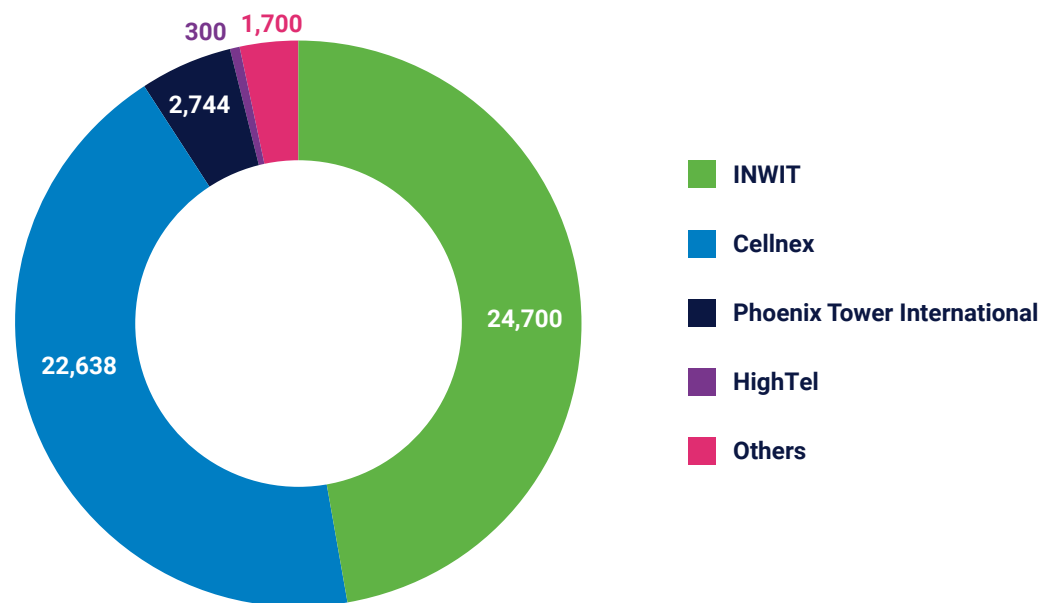


Subscribers:
80.6mn

MNOs: Iliad, TIM (Telecom Italia), Vodafone, Wind Tre (CK Hutchison)

Towercos: Cellnex, El Towers, INWIT, HighTel, Rai Way, Phoenix Tower International

Figure 21(a): Italy – estimated telecom tower ownership



Source: TowerXchange

The Italian market has seen a host of different tower monetisation strategies by its resident operators.

The first operator to monetise its portfolio was Wind (at the time owned by Vimpelcom) who carried out a sale and leaseback transaction of their portfolio of 7,377 towers (which it had carved into “Galata Towers”) to Cellnex back in 2015, originally retaining a 10% stake (with Cellnex taking over the operator’s 10% stake in 2017).

The same year, TIM, the mobile arm of Telecom Italia, carved out its tower portfolio into a dedicated infrastructure unit, [INWIT and listed a 40% stake on the Milan Stock Exchange](#), raising €875.3mn in the process.

In 2016, VimpelCom’s Wind and CK Hutchison’s 3 merged to form Wind Tre, with the merger being approved on the condition that spectrum and towers were provided to a new market entrant to ensure competition. This opened the

door to France’s Iliad who entered the market in 2018 with an aggressive low-cost strategy. In May 2019, Iliad became the second operator to carry out a sale and leaseback transaction in the country, [reaching a deal](#) to sell its 2,200 sites to Cellnex for €600mn (Iliad/Salt also agreed the sale of its French and Swiss tower portfolios to Cellnex at the same time). The deal also included a build to suit component for an extra cash consideration, with Cellnex to build up to 1,000 new towers for the operator.

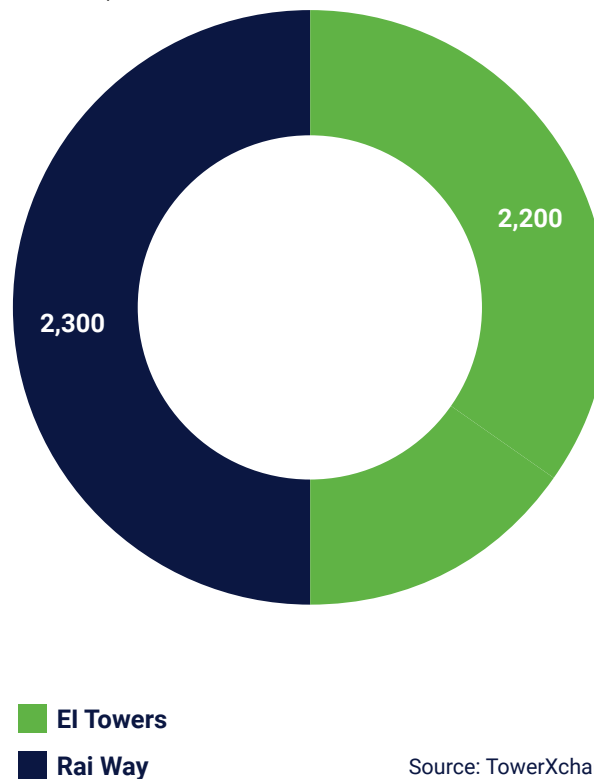
ITALY

In July 2019 Vodafone combined its 11,000 tower portfolio with TIM's INWIT (with the two operators also putting in place an active sharing agreement). Since the merger, Vodafone and TIM reduced their respective shareholdings in the towerco from 37.5% to 33.2% (with Vodafone's shareholding held by their towerco, Vantage Towers). TIM has subsequently reduced and sold remaining stakes in the holding company Daphne 3 (which in turn owns 29.9% of INWIT) to Ardian-led consortium Impulse. TIM's moves are aimed at improving cash flow.

In July 2021, INWIT reached a €70mn deal to acquire from Vodafone Italia 700 antenna systems covering road and motorway tunnels as well as 40 DAS systems covering hotels, stadiums and offices. In addition to consolidating existing infrastructure from TIM and Vodafone, INWIT has first refusal on new build for the two operators with a target of building a total of 10,000 new macro sites and small cells in its initial scope. INWIT (in conjunction with TIM and Vodafone) has also just won a government tender to deploy 5G sites in 2,000 rural locations.

In August of 2019, Hutchison announced that it planned to carve its 28,500 European towers into its a towerco unit, CKH Networks. This news was then superseded by the announcement in November 2020 that Hutchison had [signed a deal](#) to sell their European tower portfolio to Cellnex. The Italian portion of the deal added 8,900 existing sites to Cellnex' portfolio, as well as a BTS pipeline of 1,100 sites.

Figure 21(b): Italy – estimated broadcast tower ownership



Cellnex entered the Italian market in 2014, buying TowerCo (the operator which managed towers along the Italian motorway network) and its 306 sites from Atlantia. In addition to acquiring the TowerCo, Wind, Iliad and Hutchison towers, [Cellnex also acquired Italian small cells and DAS player](#), CommsCon back in 2016 and has been highly active in the small cells and DAS space.

In addition to Cellnex, there is a long tail of smaller independent towercos active in Italy. Phoenix Tower International entered the Italian market with the acquisition of Towertel from EI Towers (in conjunction with F2i). As part of the deal, PTI secured the rights to market and lease 1,600 of EI Towers' 2,300 broadcast sites. After PTI, HighTel is the next largest towerco in the Italian market with a portfolio of around 300 sites.

In early 2023 Wind Tre and Iliad announced the set-up of their 5G network sharing JV Zefiro Net that will cover 26.8% of Italian population living in rural areas. The pair will share the cost and management of mobile networks in the country's rural areas. In May 2023 Wind Tre announced the carve out and sale of its mobile and fixed network to EQT Infrastructure for an enterprise value of EUR 3.4bn. A year later the deal was cancelled due to separate network agreements with "with relevant third parties" according to Hutchinson. In addition to EI Towers having a portfolio of 2,200 broadcast sites, Rai Way too owns an almost identically sized portfolio of broadcast assets.

Vodafone has completed the sale of its Italian business to Swisscom for EUR8bn as part of plans to streamline European operations. Swisscom intends to merge Vodafone Italia with Fastweb, Swisscom's Milan-based subsidiary, a move aimed at strengthening its presence in the country and pushing fixed-mobile convergence. The deal will complete the transition of Fastweb from MVNO to national MNO and the mobile market will continue to have four MNOs.



Kazakhstan

Towers:
8,350

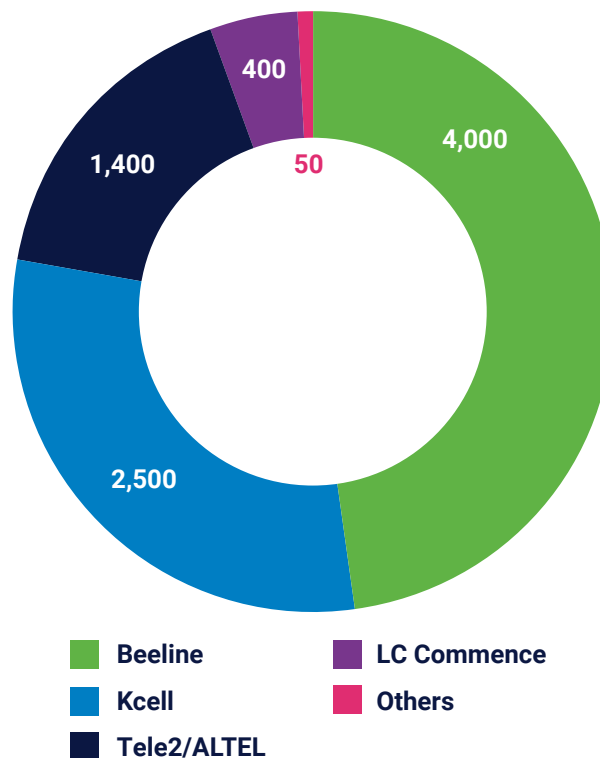
Population:
19mn

Subscribers:
25.17mn

MNOs: Beeline (VEON), kCell (Kazakhtelecom), Tele2/ALTEL (Kazakhtelecom)

Towercos: LC Commerce

Figure 22: Kazakhstan – estimated tower ownership



Source: Towerxchange
More updates on rooftops will be shared in next edition

are at present 100% owned by state-backed Kazakhtelecom. Kazakhtelecom acquired Tele2's shareholding in mid 2019 and has just announced it'll sell it to Qatari investor Power

International Holding for US\$ 1.1bn.

Both Kazakhtelecom owned MNOS won 5G spectrum licences which also include them activating over 7,000 base stations and providing 5G coverage in 75% of cities by 2027. Until recently local MNOs have considered passive infrastructure as a source of competitive advantage and have refused to share towers. TowerXchange understands that the country's MNOs will relinquish the majority of their towers.

Kcell has 2,500 towers, but TowerXchange understands it is further from selling its towers than other operators. Meanwhile, VEON-backed Beeline is believed to have a macro footprint of around 1,500 towers (in addition to shareable and non-shareable rooftops) but will eventually sell its towers to help fulfil its strategic plans to be an asset-light digital operation. It plans to create a special purpose company to house the towers and "transform" all the tower sites to get the best return on a sale.

Beeline's tower carve-out and sale strategy is becoming ever clearer in Central Asia, in fact. Over in neighbouring country Uzbekistan, the mobile operator recently established National Tower Infrastructure, which has received the majority of Beeline's tower assets, comprising approximately 4,000 towers, and is in the process becoming the leading company in the country.

Malta

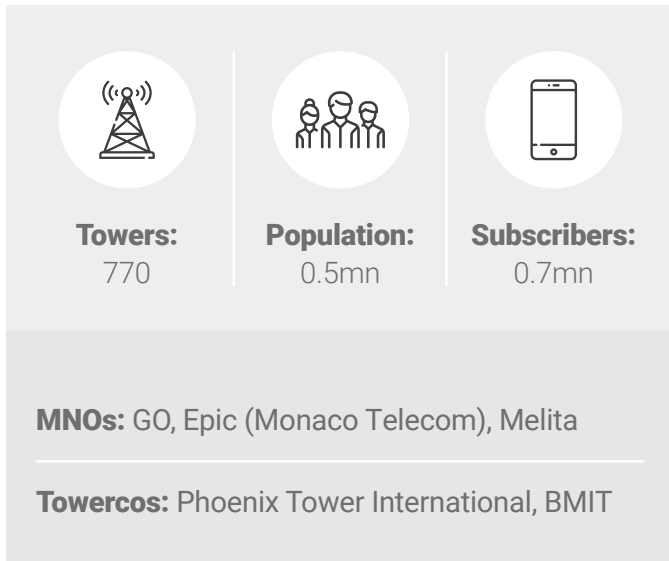
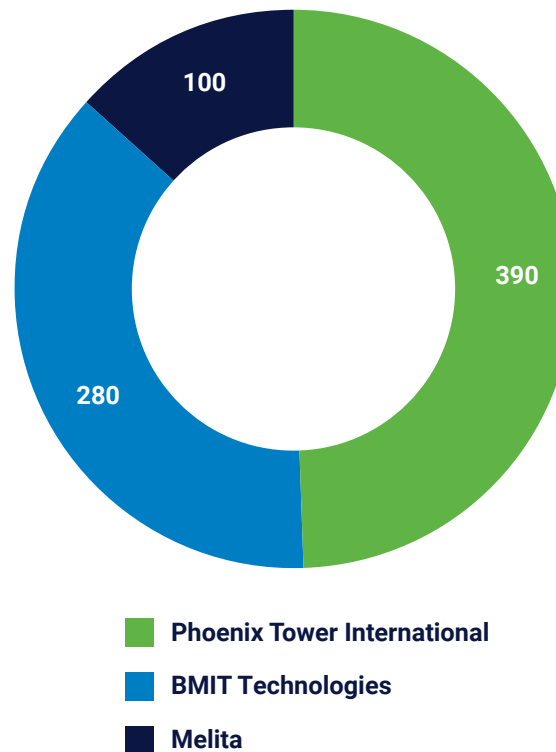


Figure 23: Malta – estimated tower ownership



Source: TowerXchange

In March 2021, Phoenix Tower International announced its acquisition of Epic's passive infrastructure which was part of a sale and leaseback deal covering the countries

of Malta and Cyprus. It included a 20-year leaseback agreement for the MNO as well as a build-to-suit programme for an undisclosed number of sites.

Epic has received EUR20 mn funding from the European Investment Bank (EIB) to accelerate the roll-out of its 5G network to reach nationwide coverage by end 2024 and to expand its fibre optic network.

GO is the second largest mobile operator with around 34% market share. In September 2023 it sold its passive infrastructure to local IT firm BMIT Technologies. The deal valued the 280 sites at EUR47.1 million (USD49.8 million) and included the installation of 30 more sites by 2030 with GO as the anchor tenant.

Melita, which was sold to EQT Infrastructure in 2019 for an undisclosed amount, has been re-sold to Goldman Sachs Alternatives.

Malta is also home to mobile virtual network operators (MVNOs) including YOM and Smarty that lease network access from the main MNOs.

Three MNOs – GO, Epic and Melita provide connectivity across the Maltese islands of Gozo and Malta.

Epic leads by number of customers with approximately 42% market share.

Formerly Vodafone Malta, it rebranded to Epic after it was bought by Monaco Telecom. The company, which is owned by the Principality of Monaco and French billionaire Xavier Niel's firm NJJ Holdings, purchased Vodafone's operations in the country in 2020.

Netherlands

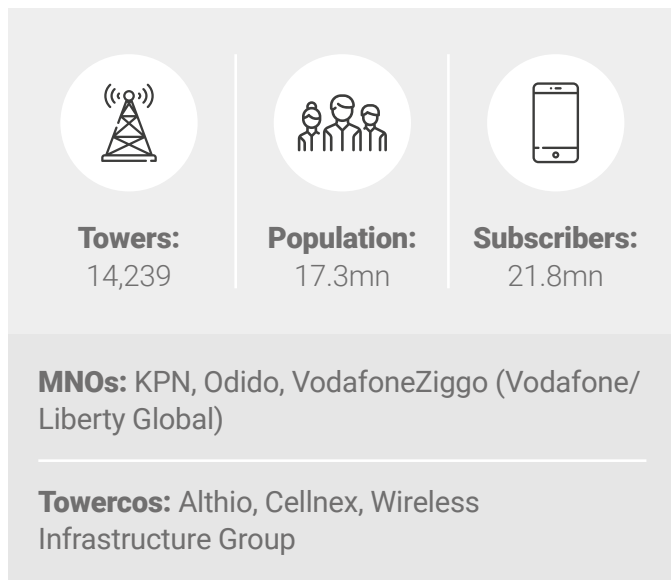
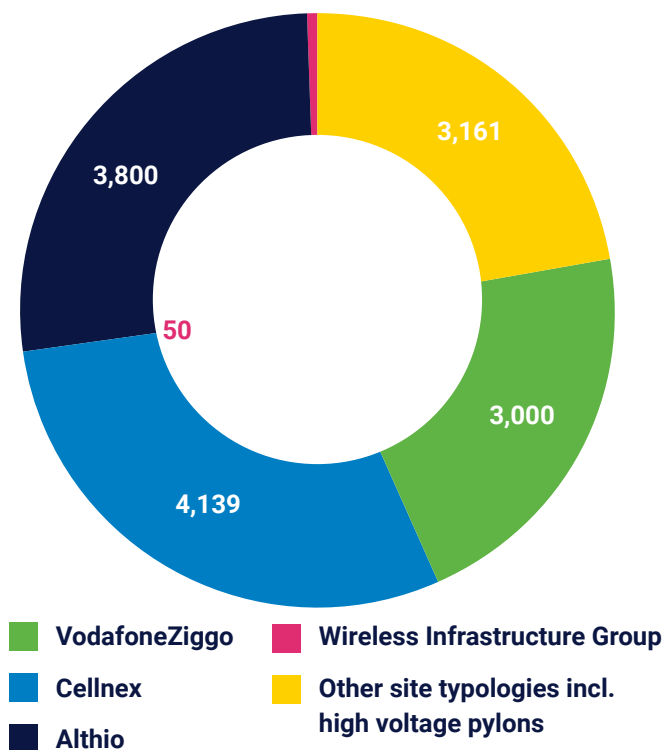


Figure 24: Netherlands – estimated tower ownership



Source: TowerXchange

since been acquired by Cellnex, who entered the market in 2016 with both deals. Cellnex also acquired Alticom and their 30 long range sites (tall towers with large equipment capacity).

Open Tower Company (OTC) which is 75% owned by Dutch pension fund ABP, and owned just over 900 macro and broadcasting towers, whilst also having access around 750 electricity pylons merged its assets with KPN's non-divested towers and formed Althio. The new towerco launched in February 2025 and manages approximately 3,800 sites representing the consolidated passive infrastructure OTC, and KPN's rooftops, macro sites and third-party towers.

In January 2021, Deutsche Telekom and Cellnex [reached a deal](#) to merge their Dutch tower businesses into Cellnex Netherlands BV whilst both becoming anchor investors of a new investment fund - Digital Infrastructure Vehicle. Digital Infrastructure Vehicle holds a 38% stake in Cellnex Netherlands as a seed asset (with Cellnex holding the remaining 62% stake). Following closure of the deal, Cellnex has a BTS pipeline of just over 200 sites. VodafoneZiggo, the final operator in the Dutch market retains its portfolio of towers, understood to number around 3,000 sites. Whilst Vodafone has carved out the vast majority of its European towers into Vantage Towers, VodafoneZiggo is a joint venture with Liberty Global.

Towerco Wireless Infrastructure Group owns a portfolio of 50 sites in the country. Additionally, there are a further c. 6,000 other locations owned by third parties that are in use as telecom sites.

There are three MNOs in the Dutch market; KPN, VodafoneZiggo (owned by Vodafone and Liberty Global), and T-Mobile (which Deutsche Telekom and Tele2 sold to a consortium of investors).

The Netherlands holds the accolade of having hosted one of the first sale and leaseback transaction in the European market, with KPN divesting towers in four tranches starting in 2008. KPN sold 601 towers across two deals to Open Tower Company, 460 sites to Shere Group and 261 sites to Protelindo. [Protelindo's Dutch portfolio](#) and [Shere Group](#) have



Norway

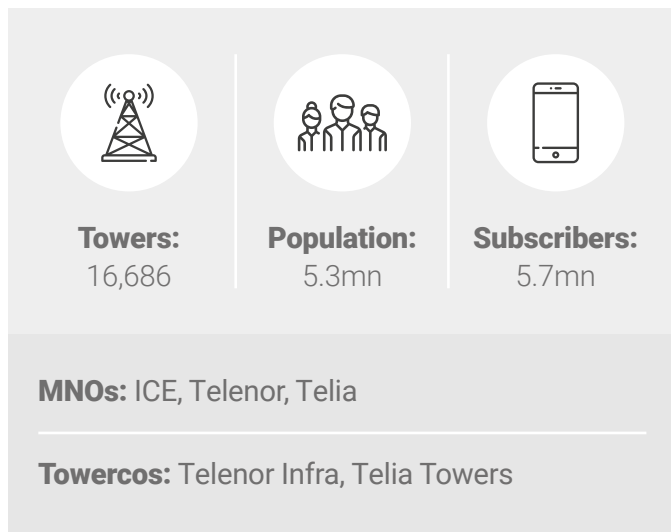
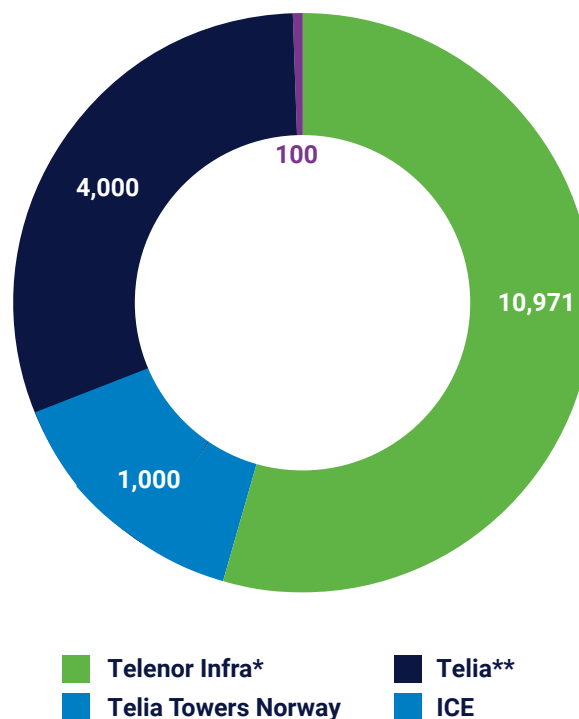


Figure 25: Norway – estimated tower ownership



*excludes Telenor Infra's c. 5,000 non-mobile sites

** Telia's ground based towers have been carved into Telia Towers Norway, they retain their rooftop sites

Source: TowerXchange

Telia had also followed in a similar strategy, carving out their tower assets in Sweden, Finland and Norway into separate towerco units. In the case of Telia, only the operator's ground based towers were carved into the towerco units, with rooftop sites remaining outside of the towerco perimeter.

In June 2021, Telia announced that [it had reached an agreement](#) to sell a 49% stake in its Norwegian and Finnish towerco units to infrastructure investor, Brookfield, and Swedish pension fund, Alecta. A similar deal shortly followed in Sweden. Telia Towers Norway is understood to own around 1,000 towers in the country, with Telia owning an estimated 4,000 further rooftop sites.

ICE is Norway's newest market entrant (having gained spectrum in 2013 and acquired Network Norway from Telia in 2015) and whilst registering just 10% of the country's subscriber base, is ambitious and poised to challenge the duopoly that has long existed in country, with the aim of reaching 25% in the longer term.

ICE has just a modest tower portfolio (estimated to be around 100 sites), with the operator relying on co-locations as well as a recently renewed roaming agreement with Telia (although continuing to build out Norway's third network).

Poland

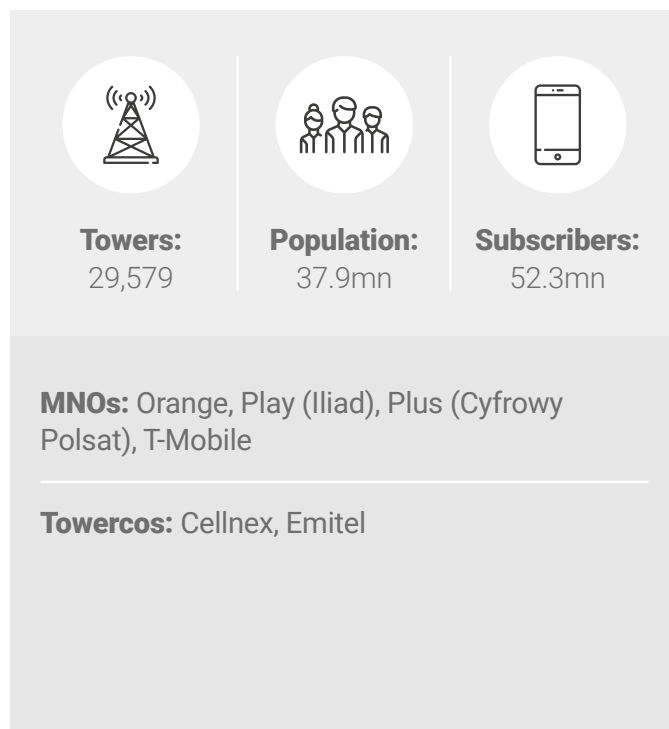
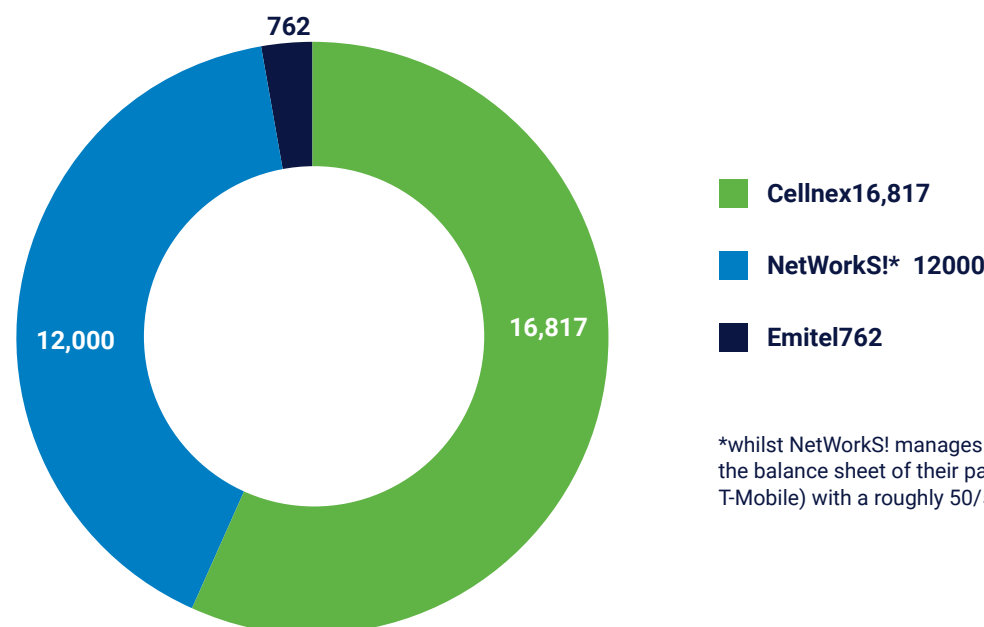


Figure 26: Poland – estimated tower ownership



*whilst NetWorkS! manages the portfolio, the towers remain on the balance sheet of their parent companies (Orange and T-Mobile) with a roughly 50/50 split

Source: TowerXchange

T-Mobile and Orange share passive and active infrastructure in Poland through 50-50 joint venture NetWorkS! While NetWorkS! operates around 12,000 towers, the assets are on T-Mobile and Orange's own balance sheets.

T-Mobile is 100% owned by Deutsche Telekom. In 2022, the operator agreed the sale of a 51% stake

in their GD Towers business covering Austria and Germany to DigitalBridge and Brookfield. Whilst the operator has sold towers to Cellnex in the Netherlands, and carved out asserts in the Czech Republic, the Polish assets remain untouched.

Orange has carved out its French and Spanish

towers into the TOTEM towerco which became operational in Q4 2021. Following the operational launch in France and Spain, Orange is looking to expand TOTEM's perimeter to other European markets, although the tie up of their Polish towers into a joint venture with Deutsche Telekom complicates carve out plans in the country.



POLAND

Independent towercos also operate in the Polish market. In October 2020, Cellnex announced that it had [reached a deal with Play](#) to acquire a controlling 60% stake in the operator's 7,000 site tower portfolio (with Iliad, Play's new owners, retaining a 40% stake - this stake has subsequently been reduced to 30%). In June 2023 Cellnex bought Iliad's 30% stake in the jointly held OnTower Poland towerco for EUR510mn reaching full ownership.

In February 2021, Cellnex announced a second deal in Poland, reaching an agreement to acquire the

network assets from Polkomtel (which trades under the Plus brand).

The deal not only includes 7,000 towers (and a BTS pipeline to roll out 1,500 new sites over the next 10 years) but also includes the acquisition of active infrastructure (c. 37,000 radio carriers, c. 11,300km of fibre backbone and FTTH backhaul and a nationwide network of microwave radio links).

The company has recently been rumoured to have

halted plans to for expansion and monetisation in the Polish market by selling a stake in its Polish business, a move that could have valued the unit at up to EUR 4bn (US\$4.4bn).

American Tower quietly entered the country in 2020 with a small-scale acquisition of just over 50 sites in the country. In June 2023 it announced its exit selling all towers to broadcast towerco Emitel for an undisclosed amount. Emitel which manages a portfolio of 677 sites, is now owned by Cordiant Digital Infrastructure.



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Portugal

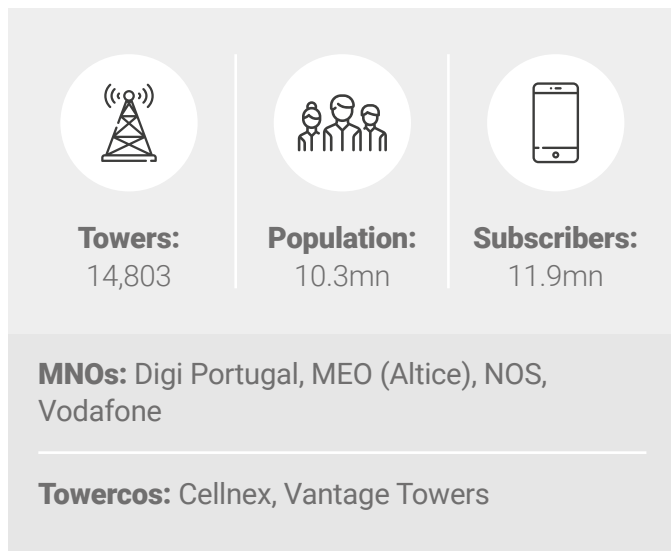
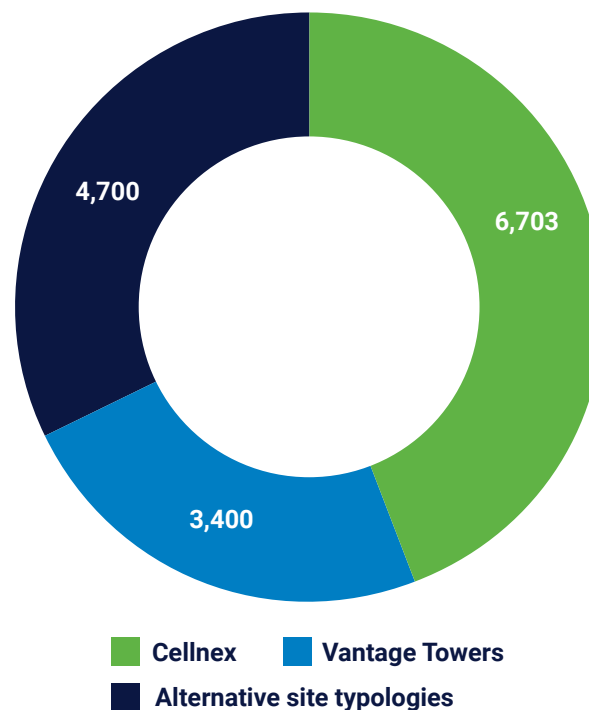


Figure 27: Portugal – estimated tower count



Source: TowerXchange

All three of Portugal's MNOs have either carved out of sold tower portfolios in the country. Altice was the first operator to execute upon its tower strategy, carving out its c. 3,000 towers into its own towerco, OMTEL (formerly Towers of Portugal). In 2018, Altice sold a 75% stake in OMTEL to a consortium including Morgan Stanley and Horizon Equity Partners. In January 2020, [Cellnex acquired 100% of OMTEL](#) for €800mn, a deal which also includes a BTS commitment of 400 new sites for MEO over the next four years. In April 2020, Cellnex signed a second deal in the Portuguese market, [reaching an agreement with MNO Nos](#), to acquire its portfolio of 2,000 sites for €375mn. The

deal also includes a BTS commitment of up to 400 new sites over the next six years. Cellnex has subsequently signed a further deal in the Portuguese market, acquiring 223 macro-sites and 464 small cell sites from Altice..

Vodafone's Pan-European towerco Vantage Towers became operational in summer 2020 and owns a portfolio of around 3,500 sites in the country. In October 2020, Vodafone and Nos reached a network sharing agreement governing both passive and active infrastructure, with the depth of sharing dependent on population density. In Q1 2023 Vodafone completed the sale of its 81.7% stake in Vantage Tower to investment firms KKR and GIP, forming a co-control partnership with them called Oak Holdings.

Spanish operator, MASMOVIL entered the Portuguese market in late 2019 through subsidiary Nowo, and secured spectrum in the much delayed 5G spectrum auction in 2021. Alongside Portugal's three established MNOs, Digi also secured spectrum and has since been developing its network for 5G. In end-2023 Digi entered into a framework agreement with Vodafone Portugal to receive spectrum usage rights and access to Vodafone's optical fibre network. In August 2024 it bought Nowo and in November 2024 launched as the fourth MNO.

Altice is reportedly looking for buyers for its Portuguese business. Telefonica, Orange and major private equity firms have reportedly been approached regarding the sale.

Romania

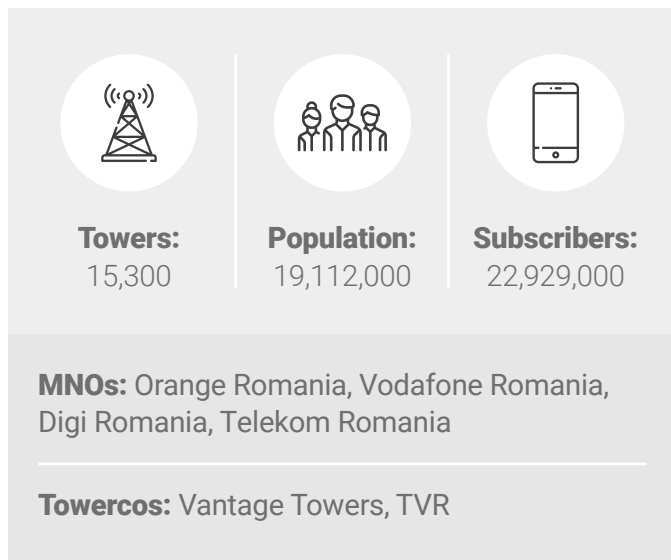
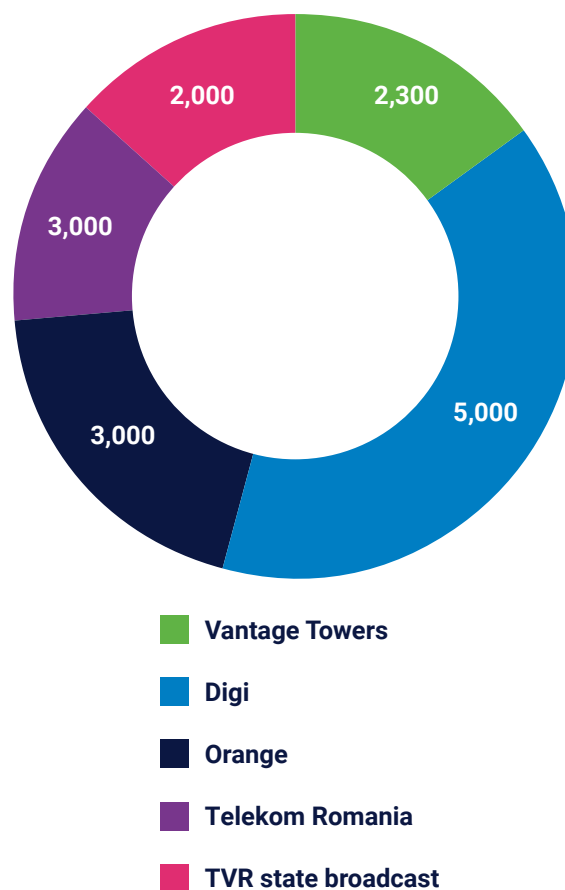


Figure 28: Romania – estimated tower count



Source: TowerXchange

There are four MNOs in the Romanian market. Orange Romania has the highest share of around 38% and with estimated 3,000 towers. It manages both active and passive infrastructure and continues building its own fibre network. In January 2024 it announced it's merging its mobile and fixed operations. Vodafone Romania is second in terms of market share. The country is one of the first towerco markets in CEE thanks to the creation of Vantage Towers. There is already a culture of network sharing, with Orange Romania and Vodafone Romania agreeing a deal to last from 2014 to 2030.

Digi Romania, part of Digi Communications, is third in terms of market share. It also leads the fixed broadband services market, supported by its strong foothold in fibre. Digi have their own passive infrastructure and own at least 6,000 towers in the country.

Telekom Romania has the smallest market share (approximately 3.8 million customers) and is wholly owned by the Greek OTE, which in turn is controlled by Deutsche Telekom. In May 2024 it signed an agreement in principle to be bought by investment company Quantum Projects Group, majority owned by Digi Communications. Digi has said once the deal is complete "Telekom Romania would remain "in the market as an independent telecommunications operator.

Digi Communications is also active in Spain and Italy, Belgium and Portugal. Digi exited the Hungarian market in 2022, selling its subsidiary to 4iG for €625 million. In line with meeting 5G coverage obligations, Vantage Towers are building new sites slowly for Vodafone, but most construction in the market is being done in-house by Digi. All tower builders in the market have had issues with steel supply due to the shutdown or redirection of Ukrainian steel production.



Russia



Towers:
154,469



Population:
144.4mn



Subscribers:
239.8mn

MNOs: Beeline (VEON), MegaFon, MTS, Tele2 Russia (Rostelecom)

Towercos: First Tower Company, New Towers, Service Telecom, Tower Infrastructure Company (plus several hundred smaller players)

activity with the announcement that it had agreed the sale of its 15,400 site Russian tower portfolio to Service Telecom. The transaction included just the operator's ground-based towers and tall shareable rooftop sites. VEON's smaller, non-shareable ones were excluded from the deal. The transaction has given significant scale to Service Telecom which had previously amassed a portfolio of 3,000 sites through a combination of new build and a handful of towerco acquisitions (Link Development in 2018, Technoresource in 2019 and Neva Telecom in 2021). Service Telecom's site count now stands at 18,396 sites and the towerco is reportedly gearing up for an IPO (although macroeconomic developments are likely to push timelines on this).

The deal between VEON and Service Telecom set of something of a chain reaction. In October 2021, the shareholders of Russia's largest MNO, MTS, approved the spin-off of MTS tower assets into a separate wholly owned subsidiary, Tower Infrastructure Company. Of Russia's operators, MTS had always been the least keen to divest towers but a few years ago announced that they were taking steps to share their infrastructure, making around a third of their total tower portfolio available for co-location.

The extent to which this had got off the ground is unclear, but the formation of Tower Infrastructure Company

places a targeted focus on driving up co-locations of their portfolio of assets, signifying a major step for the operator. Whilst MTS had plans to monetise Tower Infrastructure Company (with negotiations having reportedly been underway with players including Service Telecom and Kismet Capital), such plans are now on hold indefinitely due to the political and macroeconomic situation.

Hot the heels of MTS' announcement came an announcement from MegaFon. MegaFon had previously carved out their 17,000 towers into a separate business unit, First Tower Company with the unit having been headed by the new Tower Infrastructure Company CEO Alexander Teremetsky. In late October MegaFon revealed new plans. The operator had reached an agreement with Kismet Capital Group, the main shareholders of Russian towerco Vertical. Under the terms of the agreement, Megafon will transfer ownership of its 17,000 towers to a new towerco venture, securing a 25% stake in the entity. Vertical will contribute its 5,000 sites to the joint venture, with Kismet Capital securing a 65% stake in the entity.

Just a couple of months later, news emerged that the new Kismet Capital owned towerco, New Towers, had reached a deal to acquire Russian Towers. One of Russia's most notable towercos, Russian Towers had built up a portfolio of 7,500 sites through both new

Sanctions imposed by the international community on Russia are a dominating factor in the market dynamics with mobile and tower activity recently focused on international entities exiting the market. Veon exited Russia in Oct 2023 by closing the sale of its Russian operations to a group of senior members of the PJSC VimpelCom management team, led by VimpelCom CEO Alexander Torbakhov.

In September 2021, VEON kicked off the wave of local

RUSSIA

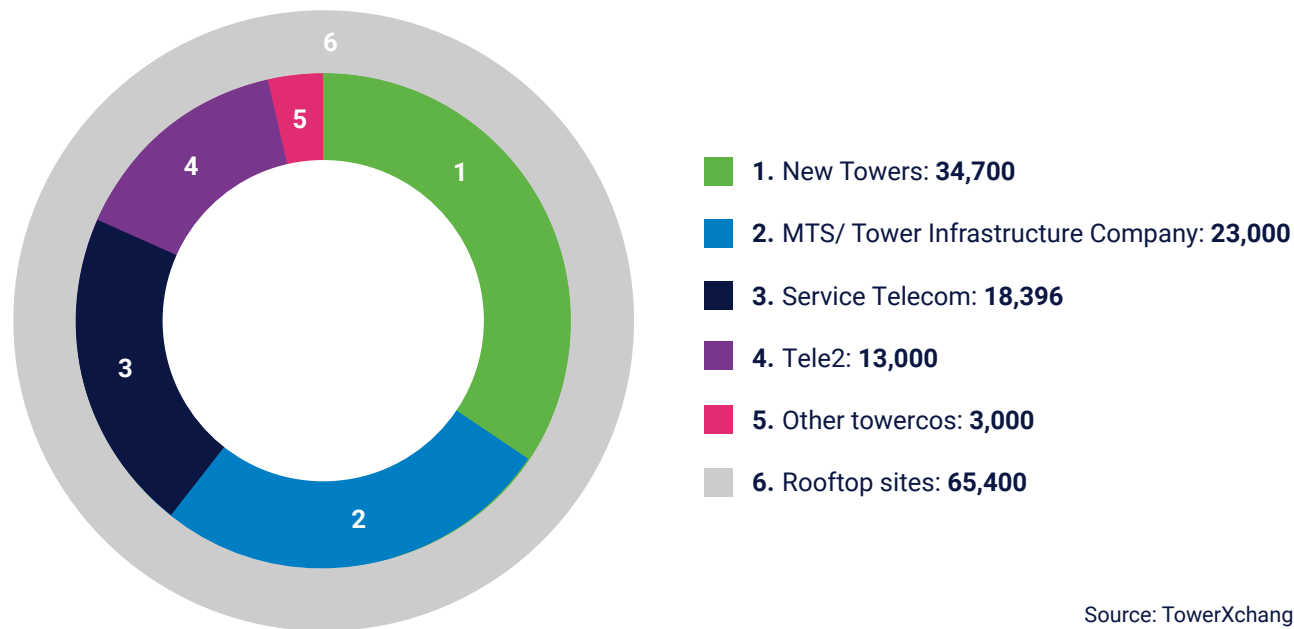
build and the rollup of smaller towercos in the market. The Russian Towers' team bring national towerco experience to New Towers, with Vertical's footprint having previously been confined to the Moscow region.

Tele2, Russia's newest mobile market entrant which was acquired by Rostelecom in 2020 is continuing to roll out its network and has a portfolio of around 13,000 sites. Prior to the takeover by Rostelecom, Tele2 had announced that it was exploring a sale of its assets. Whilst no further details have been announced since the takeover, it was widely expected that Tele2 would follow in the footsteps of its peers, carving out or selling its tower portfolio.

Beyond Service Telecom, New Towers and Tower Infrastructure Company, a fragmented ecosystem of smaller players collectively owns around 3,000 sites in the country. In addition to Russia's c. 85,000 ground based towers, there are a 65,400 rooftop sites in use across the Russian market. In terms of 4G coverage most work has been done, bar road and rail networks where coverage has been limited to date. Beyond extending coverage to such transport routes, the main driver of new site rollout is improved capacity in dense urban environments.

5G rollout will present further organic growth opportunities. A lack of suitable spectrum has been holding back the technology with President Putin having reserved the most popular 3.4-3.8GHz band for

Figure 29: Russia – estimated tower count



military and intelligence uses. In 2021, Russia's four MNOs announced the formation of a joint venture "New Digital Solutions" which aims at taking a collaborative approach in testing and pushing for the release of new frequency bands suitable for 5G networks. The government had previously proposed auctioning 5G spectrum to a single operator, thus creating a single 5G network but operators in the market had been sceptical.

Another feature of 5G discussions in the Russian market is the government's specification that

operators must use domestically manufactured equipment. As a result, Nokia, Ericsson and Huawei began setting up production facilities in the country, although sanctions have impacted further development. Local conglomerate, Ros Technologies is involved in the design and manufacture of their own equipment. In mid-2021, Russia's Deputy Prime Minister Dmitry Chernyshenko announced that 5G networks would be launched in the largest cities in the next three years suggesting a 2024 commercial launch of the technology.



Serbia

Towers:
5,496

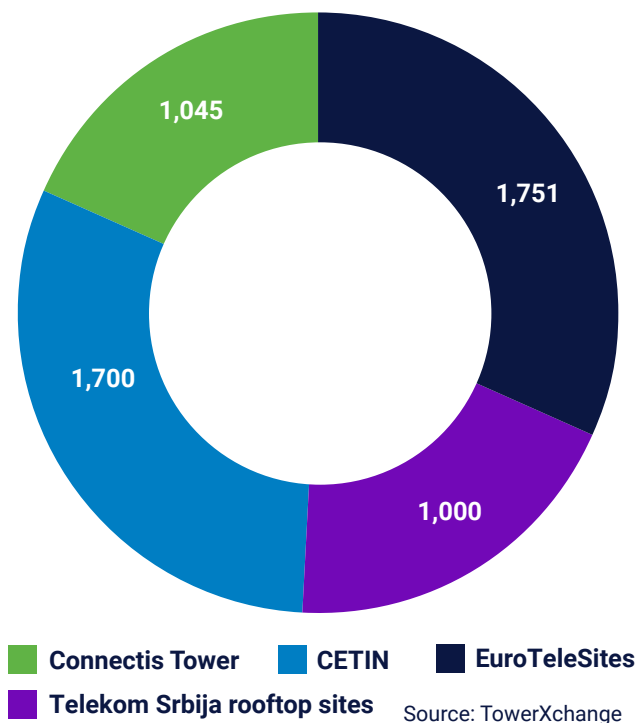
Population:
6.9mn

Subscribers:
8.5mn

MNOs: mts (Telekom Srbija), Yettel (PPF Group), A1 (Telekom Austria)

Towercos: EuroTeleSites, CETIN, Connectis Tower

Figure 30: Serbia – estimated tower ownership



In July 2020, Yettel (owned by PPF Group, recently rebranded from Telenor Serbia) followed in the footsteps of PPF's Czech opco (O2 Czech Republic) and separated out their retail and infrastructure divisions, with their infrastructure carved out into CETIN. CETIN's remit extends much beyond towers, with CETIN taking ownership of the operator's metallic cable pairs, fibre and indoor and outdoor base stations. In Q4 2021, PPF announced the sale of a 30% stake in CETIN to Singapore's GIC investment fund. In Q3 2023, the PPF Group announced it has agreed the sale of a 50% plus one share in the CETIN as well as its MNO unites

in Bulgaria, Hungary, and Serbia to UAE-based e&. The deal is being investigated by the European Commission.

MNO A1 which has rebranded from vip, is owned by Telekom Austria Group. A1 Telekom Austria Group carved out their tower portfolios in a number of markets,

including Serbia, into dedicated towerco units going under the EuroTeleSites brand. Owned by ÖBAG and América Móvil, it was set up following the completion of Telekom Austria Group's tower spin off in Austria in early 2023.

mts (owned by Telekom Srbija has Serbia's largest subscriber base with a 44% market share.

In May 2023, Serbia enacted its updated Electronic Communications Code which now aligns with the EU's Electronic Communications Code and incentivises the deployment and use of new network technologies, and the development of electronic communications. Serbia's much awaited 5G spectrum auction is expected to take place in late 2025 which would present monetisation opportunities for local market players. They've already started to roll out 5G infrastructure.

In January 2024 mts's owner Telecom Srbija completed a tower sale process covering sites owned in Serbia, Bosnia and Herzegovina and Montenegro to a consortium led by infrastructure investor Actis. The Connectis Tower towerco has since expanded Serbian macro tower-only portfolio with rooftop sites by acquiring assets from Dot Networks.



Spain



Towers:
40,041



Population:
47.1mn

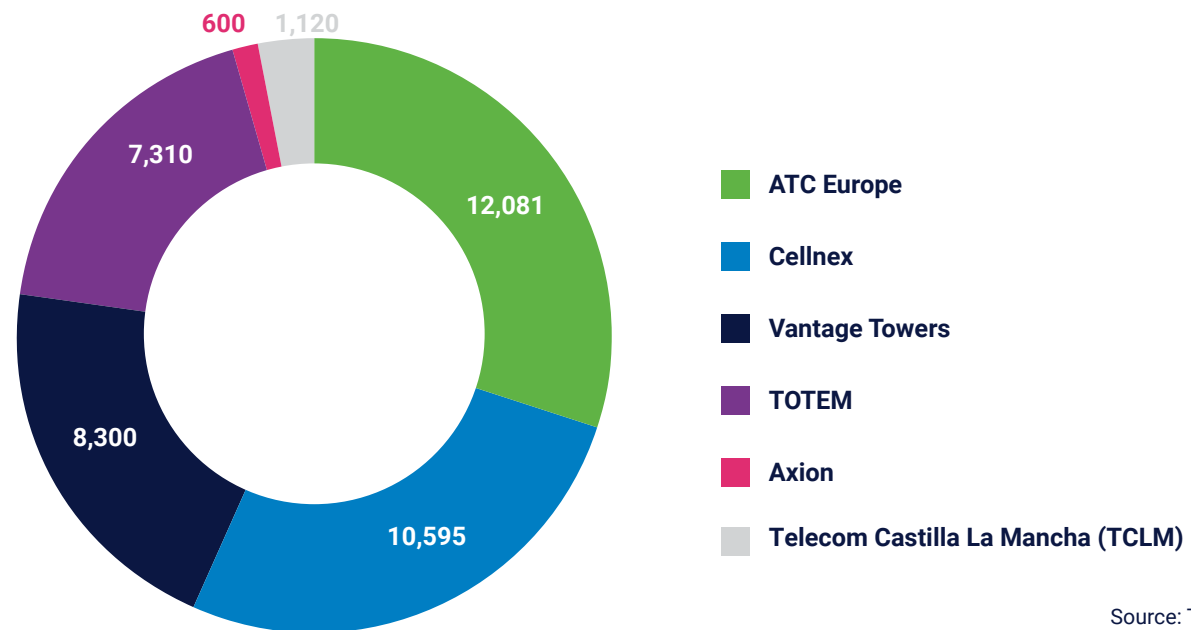


Subscribers:
55.3mn

MNOs: Movistar (Telefónica), MasOrange, Vodafone (Zegona Communications)

Towercos: ATC Europe, Axion, Cellnex, TOTEM, Telecom Castilla La Mancha (TCLM), Vantage Towers

Figure 31: Spain – estimated tower count



Source: TowerXchange

100% of Spain's towers are now owned by towercos.. Spain is the headquarters of multinational towerco, Cellnex and is where the company began. Cellnex acquired the bulk of its Spanish telecom tower portfolio through a number of transactions with Telefónica and Yoigo across 2012-2014 (and also owns a portfolio of broadcast assets in the country).

Telefónica [carved out the majority of its Spanish towers](#) into infraco, Telxius, back in 2016 (transferring the remainder of its sites in Q3 2019). In January 2021, American Tower announced that it had [reached a deal](#) to acquire Telxius' tower division, giving the towerco a new European market to its operations. The deal has now been closed with the towers being moved into ATC

Europe – American Tower's European arm in which CDPQ and Allianz have acquired a 48% stake (PGGM, ATC Europe's JV partner prior to the Telxius deal has restructured its stake in ATC Europe to a shareholding in the local German and French tower units).

Vodafone sold an 81.7% of its stake in its European



SPAIN

toweco Vantage Towers to a consortium of investment firms KKR & Co., Global Infrastructure Partners (GIP) and The Public Investment Fund of Saudi Arabia (PIF). The co-control partnership of the MNO, GIP and KKR called Oak Holdings has bought out the minority shareholders in Vantage Towers and now holds a 89.3% stake. The towerco was delisted in end 2023.

Orange announced the operational launch of TOTEM in October 2021, with the operator having carved out its French and Spanish assets into its towerco unit. In Spain, TOTEM's portfolio is roughly 50/50 split between ground-based towers and rooftops. TOTEM's perimeter is initially confined to France and Spain but the operator is undertaking a strategic review of its other markets. Orange has stated that it plans to retain control of TOTEM but is open to exploring the formation of joint ventures with other MNO owned towercos. Vantage Towers operates in the Spanish market and so could be one joint venture under consideration with Vantage Towers having combined its Italian towers with those of TIM in INWIT. Vodafone and Orange already have an active sharing agreement in place in Spain.

In 2019, Orange sold 1,500 non-strategic sites to Cellnex.

In terms of preparation for 5G, Spain's towercos have been proactive, with Cellnex announcing some high profile small cell and DAS partnerships with venues such as Atlético de Madrid football club's new Wanda Metropolitano and partnering with JCDecaux, and Axion partnering with UK towerco Wireless Infrastructure Group to form a joint venture

called Iberia Small Cells Network, which will provide neutral host services to venues and municipalities in the region. In 2024, Vantage Towers announced a 10-year long project where it is deploying for their customer Telefonica in cooperation with TOTEM, potentially the first big wave of the outdoor small cell demand in Europe.

Both Cellnex and Axion have also entered the fibre space with Cellnex acquiring Xarxa Oberta de Catalunya (with 3,000km of fibre) and Axion partnering with Enagas to launch a new fibre operator Axent, with a national network of 5,000km across Spain. Vantage Towers recently signed an agreement with Movistar to carry out 5G upgrades at 1,500 sites

MNO consolidation in Spain is now a reality: Orange and MASMOVIL received the approval of the competition authorities of the European Commission (EC) and Spain to merge. The new MNO has been named MasOrange. MVNO Digi (part of Romanian DIGI Communications Group) has received remedies as stipulated by the EC. These include a spectrum transfer agreement from MasMovil and a national roaming agreement with Orange. These present an opportunity for Digi to build its mobile network and become Spain's next MNO.

UK-based investment company Zegona Communications has completed the purchase of Vodafone Spain for EUR 5bn (US\$5.3bn), representing a multiple of 3.9x EBITDAaL. Zegona will be able to use the Vodafone brand in Spain for up to 10 years post completion.



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Sweden



Towers:
18,048



Population:
8.6mn

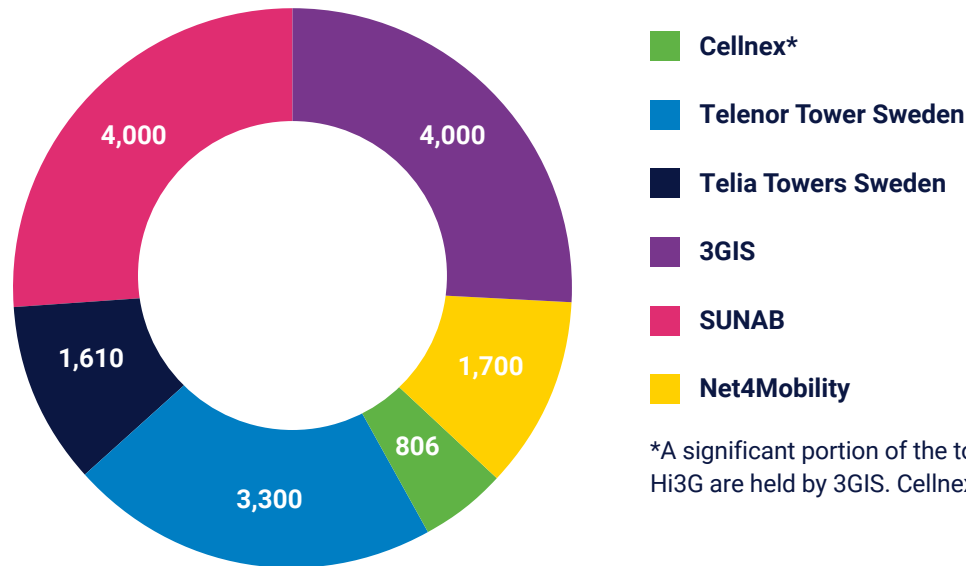


Subscribers:
12.7mn

MNOs: Hi3G (Hutchison), Tele2, Telenor, Telia, Teracom

Towercos: 3GIS, Cellnex, Net4Mobility, SUNAB, Telenor Towers Sweden, Telia Towers

Figure 32: Sweden – estimated tower ownership



Sweden is the European market with the deepest and most complex level of network sharing, with sharing in place between all four of the country's main MNOs. When Telia, the incumbent at the time, failed to secure a 3G license back in 2001, an agreement was reached with Tele2 to build a joint 3G network using Tele2's license. The 50:50 joint venture company, Svenska UMTS-nät AB (SUNAB) was formed to build, own and operate that network and involves RANsharing and shared spectrum. SUNAB manages a network of 4,000

sites. Telenor and Hi3G also signed an agreement in 2001 to jointly deploy a 3G network, with the joint venture company, 3GIS, responsible for building, owning and operating the network. Whilst the joint venture involves RANsharing, it does not involve shared spectrum. 3GIS currently owns a portfolio of around 4,000 sites (of which c. 2,700 are ground based towers and 1,300 rooftops). With 3G networks set to be switched off by 2025, the two joint ventures are set to be wound down. As to the fate of the towers owned

by the joint ventures, this is not yet clear, with them being shared between their two parent companies one possible outcome.

In 2009, the level of network sharing further intensified when Tele2 and Telenor Sweden formed a joint venture, Net4Mobility, to build and operate a nationwide 2G and 4G network together. This collaboration has extended into the 5G era, with Net4Mobility obtaining spectrum in Sweden's 5G auctions for the rollout of a joint 5G network

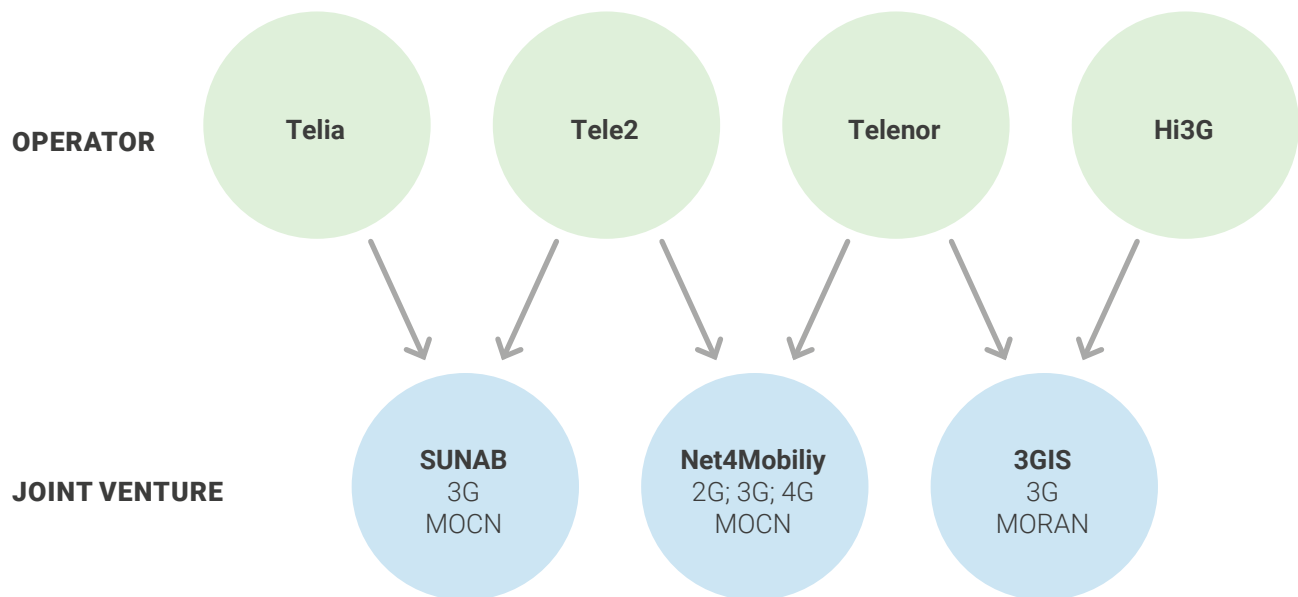
for their parent companies. The joint venture involves RANsharing and shared spectrum.

Whilst the joint venture between Tele2 and Telenor has extended its scope, Tele2 have also hinted that they may explore alternative strategies for their infrastructure (although not specifically referencing Sweden) with their CEO telling analysts “There is an opportunity to do more in terms of sharing and separating out some of our infrastructure assets” adding that they have a “very pragmatic approach to what they need to 100% ourselves and what we can share and where we can have partners.”

Telenor has carved out their Nordic assets into Telenor Tower Holdings. Whilst Telenor’s joint ventures were initially included in Telenor Tower Holdings’ perimeter, the towerco currently only reports financials for its wholly owned towers. Telenor Tower Sweden owns 1,688 towers in the market with a roughly 50/50 split between rooftops and ground based towers (and has a tenancy ratio of 2.3x thanks to the high levels of infrastructure sharing in the market). In their 2022 Capital Markets Day, Telenor announced that they plan to monetise a minority stake in one or more of their national towercos in the next 12-24 months.

Telia had also followed in a similar strategy to Telenor, carving out their tower assets in Sweden, Finland and Norway into separate towerco units. In the case of Telia, only the operator’s directly owned shareable towers

Figure 33: Sweden – network sharing joint ventures



[were carved into](#) the towerco units, with single tenanted rooftop sites and stakes in joint ventures remaining outside of the towerco perimeter.

Following on from deals in Norway and Finland, Telia [agreed the sale of a 49% stake in Telia Towers Sweden](#) to infrastructure investor Brookfield and Swedish pension fund, Alecta. Telia Towers Sweden owns 1,600 towers in the country. In May 2024 it announced a strategic acquisition of 10 towers in the south of Sweden from European Connectivity Networks.

Hi3G, owned by Hutchison, have sold their towers to Cellnex as part of a [pan-European deal](#).

Cellnex has acquired 2,650 towers from the operator in Sweden (with a certain portion held by 3GIS, Hi3G’s JV with Telenor) and expects to add up to 2,550 sites over the next eight years.

In November 2023, investment firm Stonepeak completed the acquisition of a 49% stake in Cellnex Sweden and Cellnex Denmark for EUR 730mn.

Switzerland

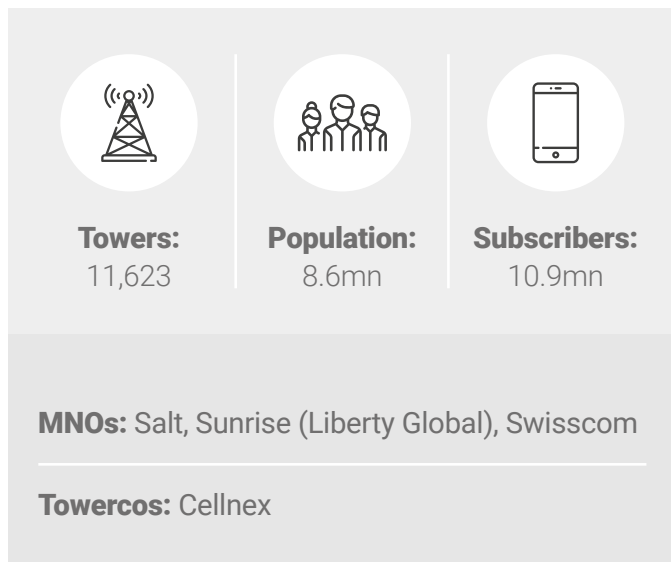
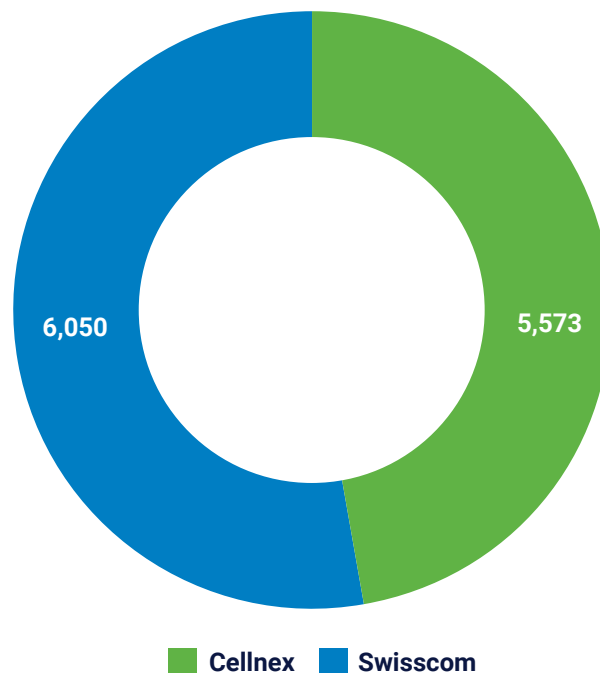


Figure 34: Switzerland – estimated tower ownership



Source: TowerXchange

towers in the Swiss market, with this figure set to rise to 6,118 once all BTS commitments are realised. It's rumoured to be considering the sale of its Swiss assets as part of a company-wide process of rationalisation and deleveraging. Swisscom is understood to have around

6,050 sites in the market.

Outside of the country, Swisscom has agreed to buy Vodafone's Italian business to for EUR8bn. The sale is part of Vodafone's wider plans to streamline its European operations. Vodafone will continue to provide certain services to Swisscom for up to five years as part of the transaction.

Around two thirds of Switzerland's cell sites are rooftops as opposed to ground based towers, although many of those rooftops have 'stub' towers on them, making them more readily co-locatable. New build (and co-location) dynamics in Switzerland are affected by EMF regulation, which are the strictest TowerXchange has encountered.

The Swiss regulator Ofcom has confirmed it is planning to make frequencies in the 3400MHz-3500MHz range available for private 5G mobile networks. At present it is working on finalising the regulatory and technical framework for the licensing of campus networks.

The Swiss market has seen two sale and leaseback transactions by its operators in recent years. In 2017, [Sunrise reached a deal](#) with a consortium involving Cellnex, Swiss Life and Deutsche Telekom Capital Partners for the sale and leaseback of its 2,399 sites for €430mn (with the deal including an additional build to suit component which also involves the deployment of 200 DAS nodes). In 2019, [Salt reached a deal](#) with Cellnex for the sale of a 90% stake in its 2,800 site portfolio (with Cellnex planning further investment to add up to 500 further sites for the operator). Cellnex now owns 5,573

Turkey

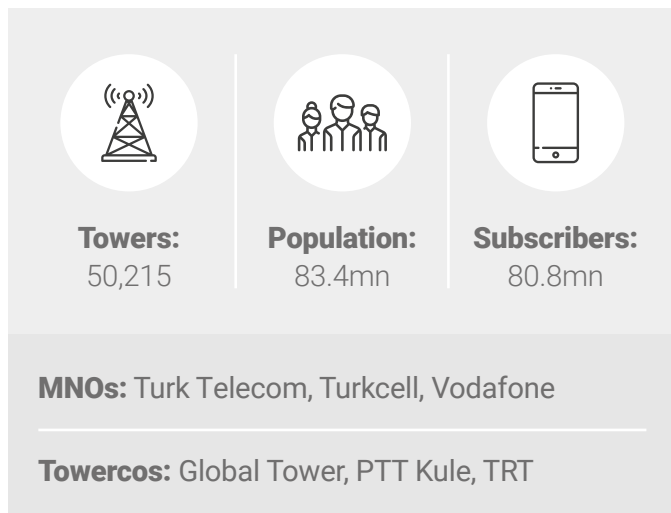
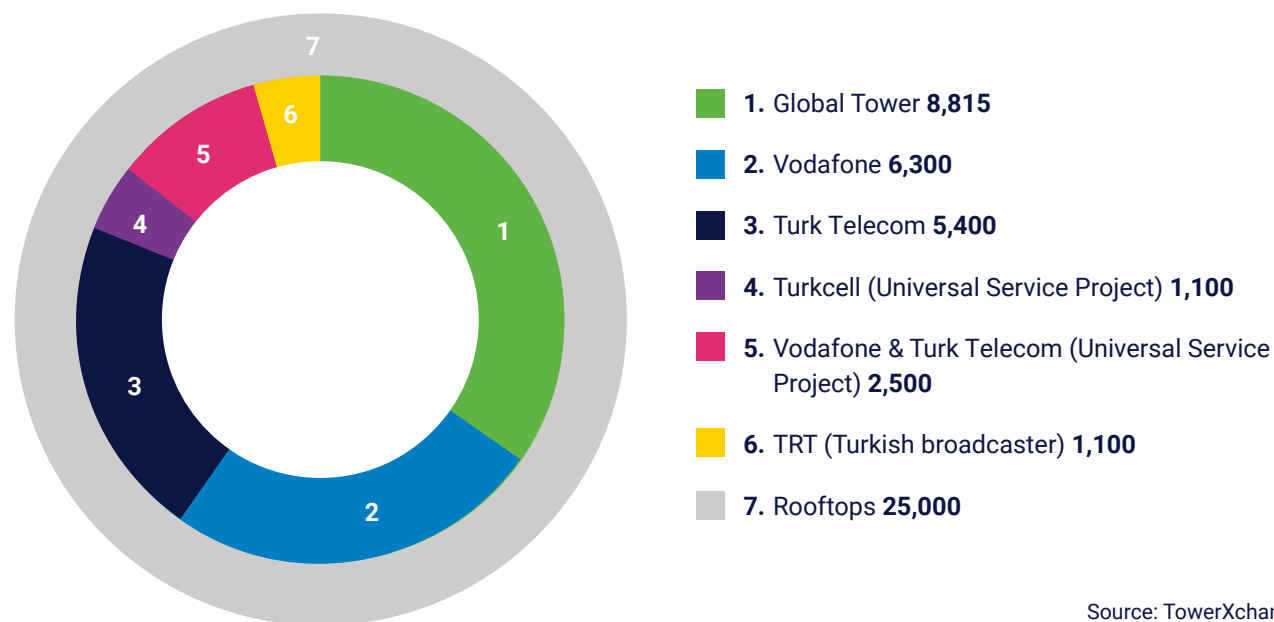


Figure 35: Turkey – estimated tower ownership



The Turkish tower market is dominated by Global Tower, Turkey's biggest towerco, currently 100% owned by Turkey's #1 MNO, Turkcell. Of the 8,815 towers marketed by Global Tower, around half are on the towerco's balance sheet, the other half remain on Turkcell's balance sheet.

Global Tower initially struggled to be recognised as independent of its parent company: for several years Turk Telecom refused to co-locate on Global Towers' sites, although Vodafone did from the outset. Turkish MNOs remain reluctant to pay a commercial rate to lease a site, resulting in Global Towers' tenancy ratio

(~1.3) being lower than other towers (~1.5) which are shared on a barter basis.

In 2016 Global Tower attempted an IPO on the Istanbul Stock Exchange, which was postponed shortly before the planned flotation date. Since pulling the IPO

they have focussed on growth, offering fibre to their customers through Turkcell's fibre arm, SuperOnline, reselling satellite connectivity to both their tenants and aiming to acquire and consolidate the majority of the towers in Turkey before embarking on a large decommissioning regime. In end 2022 Global Tower's



TURKEY

parent company Turkcell announced it has started a preparation process for an IPO of a certain amount of the towerco's shares, which will be held in Turkey, and when required on international markets, depending on the market conditions.

Whilst Vodafone has carved out its European towers into Vantage Towers, their Turkish operations straddling Europe and Asia were not included in such plans. Vodafone is understood to own around 6,300 towers in the market.

Turkey's final operator, Turk Telecom is understood to own around 5,400 towers.

A quirk of the Turkish market is that the infrastructure rolled out for 2G in the early 2000s, a total of around 16,000 towers, will revert to the Ministry of Transport and Communications in 2023. As yet it is not clear what the government will do with these assets: it has been widely believed that the government would not have the appetite or resources to manage such a large portfolio of towers and would hand them back to the MNOs, but the current investibility of telecoms infrastructure means no option is currently off the table.

Turkey is home to one of the world's largest government-owned universal service networks, called 'Universal Services Project' and implemented by the Ministry of Communication and Transportation. Phase one of the project was auctioned in 2011, with Turkcell

implementing 1,100 rural sites following a successful bid. Phase two was awarded to a joint venture between Vodafone and Turk Telekom with plans to build a further 2,500-3,000 RANsharing sites in rural areas.

In addition, the Turkish government has launched a new towerco called PTT Kule Inc, which is focussed on very large scale tower structures for telecoms and broadcast as well as hosting data centres and other facilities. PTT Kule are planning as many as 40 high towers across Turkey in the next few years.



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Ukraine

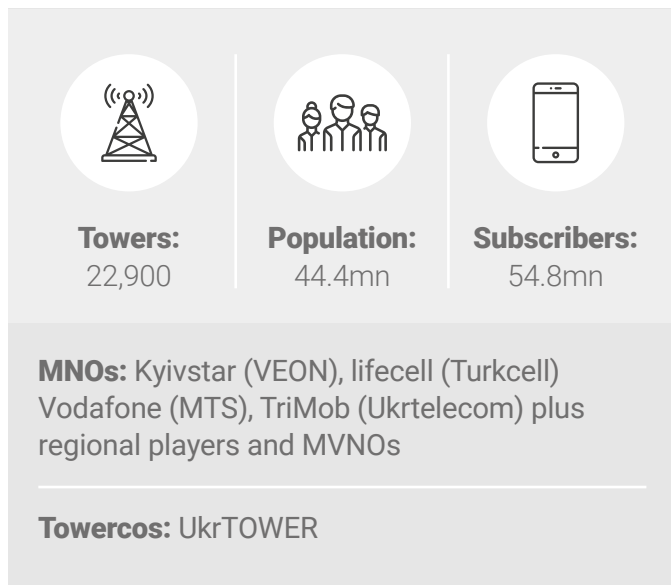
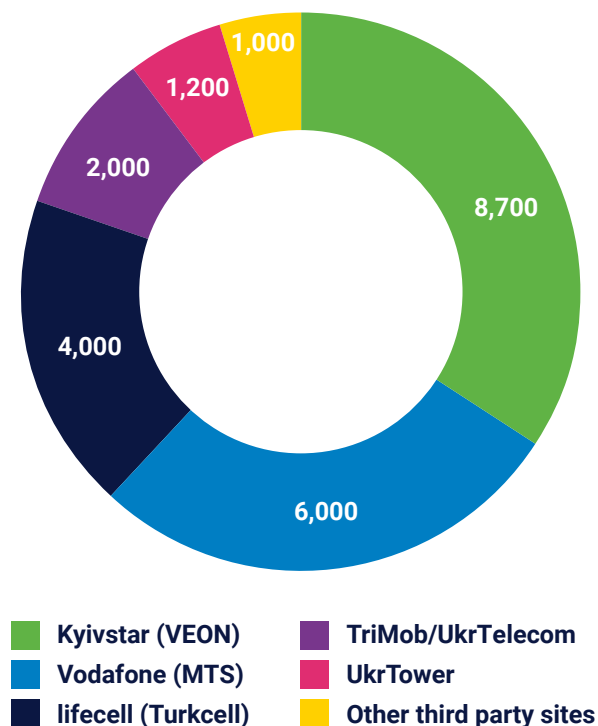


Figure 36: Ukraine – pre-war tower estimates



Source: TowerXchange

their captive towerco Global Tower. UkrTOWER's current site count is 1,200, including a number of in-building solutions. Outside of UkrTOWER, only 3-5% of towers and rooftops

in Ukraine are shared, mostly at nominal prices or under barter arrangements. In 2023 Turkcell signed an agreement to sell all shares of lifecell and UkrTOWER to Xavier Niel's NJJ Capital. The US\$524.3 mn sale (which also includes consultancy Global Bilgi) completed in September 2024.

In 2017 VEON decided to bring their Ukrainian towers to market (in conjunction with towers in Kazakhstan, Armenia and Georgia). The deal was quietly scrapped after the sale of VEON's Russian towers was cancelled in May 2017. In their Q4 2020 results, VEON announced that they were creating towerco business units in each of their nine markets, remaining open minded about with what they will do with their towerco units. In September 2021, VEON announced that it had reached a deal to sell its Russian tower portfolio to independent towerco Service Telecom. The operator has ear marked Ukraine (alongside Bangladesh and Pakistan) as its next priority to monetise, although macroeconomic events have put any developments on hold in the Ukraine.

Vodafone's site count is thought to stand at 6,000, TriMob's at 2,000 and lifecell are understood to retain around 4,000 towers in the market. A further 1,000+ leased third party structures, such as smoke stacks and water towers, are utilised by Ukraine's MNOs, while the core site count includes around 60% ground based towers, 40% rooftops, bringing the total unique site count (excluding co-locations) to around 22,800.



United Kingdom

Towers:
45,623

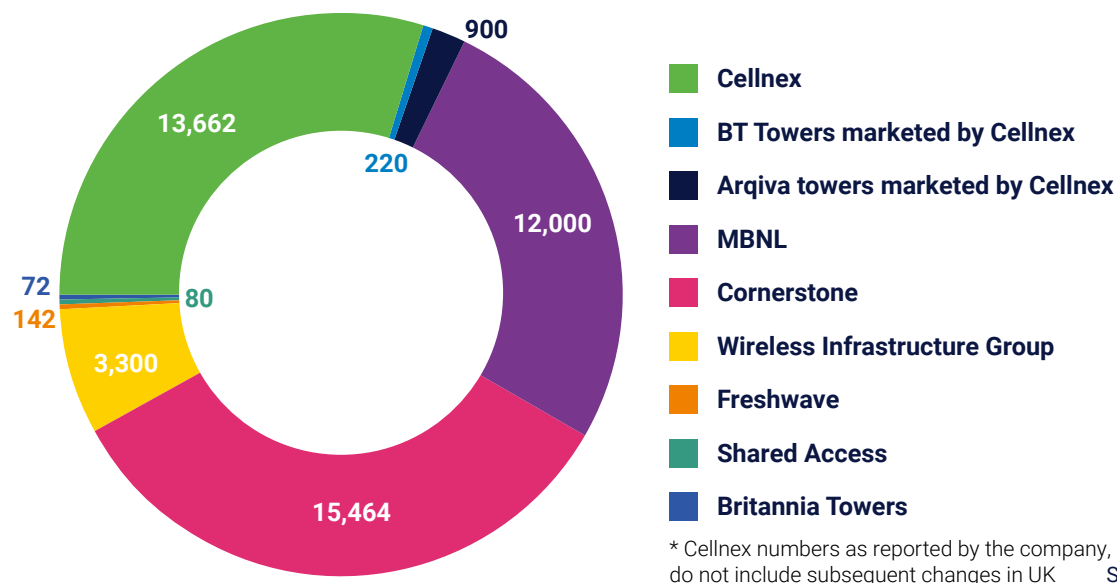
Population:
66.8mn

Subscribers:
79.5mn

MNOs: 3 (CK Hutchison), EE (BT), Virgin Media O2 (Telefónica/ Liberty Global), Vodafone

Towercos: Britannia Towers, Cellnex, Cornerstone, Freshwave, MBNL, Shared Access, Wireless Infrastructure Group

Figure 37: United Kingdom – estimated tower ownership



* Cellnex numbers as reported by the company, do not include subsequent changes in UK Source: TowerXchange

The UK market has a unique structure with two joint venture infracos, Cornerstone (between Vodafone and O2 (Telefónica)) and MBNL (between EE (BT) and 3 (Hutchison)) responsible for over 70% of the country's total stock of towers. Whilst infrastructure sharing businesses in their own right (MBNL covering both active and passive equipment, and Cornerstone just passive), Cornerstone and MBNL are the primary clients of UK's towercos managing leasing

arrangements on behalf of their own shareholders. Whilst Cornerstone has its own balance sheet, the towers managed by MBNL remain on the balance sheet of the parent MNOs. Of MBNL's 12,000 sites, around 7,500 are shared by both Hutchison and EE, with 4,500 used by EE only.

In early 2021, Vodafone and Telefónica announced the commercialisation of Cornerstone, entering into long term

MSAs with Cornerstone. Cornerstone will also become the preferred supplier of new sites for the two operators, with Telefónica and Vodafone committing to be anchor tenants on c. 1,200 new sites to be built by Cornerstone. The majority of such new site build is due to the unwinding of the active sharing in place between Vodafone and Telefónica in large urban centres (the two parties' active sharing agreement is to provide for network coverage



UNITED KINGDOM

outside of large urban centres). Vodafone has transferred its 50% interest in Cornerstone to its towerco, Vantage Towers, adding a further country to the towerco's portfolio.

Telefónica had carved out its tower portfolios in Europe (bar the UK) and CALA into its infraco, Telxius, prior to [agreeing the sale](#) of Telxius' tower business to American Tower in early 2021. Following the merger of O2 and Virgin Media, Telefónica and Liberty Global started a process to sell a stake in Cornerstone. After months of speculation, in the UK, Virgin Media sold a 16.67% stake in Cornerstone to GLIL Infrastructure for GBP360mn (US\$459mn).

In the case of MBNL, Hutchison has monetised its stake in the JV, with Cellnex set to acquire the economic risks and rewards of Hutchison's stake as part of their pan-European deal with the MNO. As a result, Wireless Infrastructure Group has acquired 1,100 towers from Cellnex UK. As part of the deal, Cellnex plans to build a further 600 new sites in the UK over the next eight years. In late 2024 the UK authorities approved the proposed merger between Vodafone and Three into a combined company where Vodafone will own 51% of the combined business with Three's parent company CK Hutchinson owning 49%. More deal-closing conditions are to be announced.

Cellnex entered the UK market in 2016, with the [acquisition of Shere Group](#) and their portfolio 540 UK sites (whilst also acquiring their Dutch assets). The towerco then strengthened their position in the UK signing a [long-term strategic agreement](#) with BT for the management and

marketing of 220 high towers owned by the operator. In October 2019, Cellnex announced that it had [reached a deal](#) to acquire the telecoms division of Arqiva. In the £2bn deal, Cellnex took ownership of 7,400 sites whilst securing the rights to market a further 900 Arqiva sites, giving the towerco significant scale in the UK market.

DigitalBridge (formerly Digital Colony) has been similarly acquisitive in the UK market, albeit more focussed on the small cells and DAS market. In 2018, Digital Colony acquired two UK indoor coverage providers, [Stratto](#) and [OpenCell](#), then in 2019 acquired [iWireless Solutions](#) and [Spyder Facilities](#). DigitalBridge has combined the four businesses into their digital infrastructure platform, [Freshwave](#).

In addition to Cellnex and Freshwave, there are a number of other independent towercos operating in the UK market, the largest of which is Wireless Infrastructure Group with over 3,200 sites. WIG also has a significant in-building portfolio of sites. In late 2019, Wireless Infrastructure Group was [acquired by Brookfield](#), giving them significant financial backing to participate in any European M&A on the horizon.

In a bid to address rural broadband coverage in the UK, the country's four MNOs have signed up to the Shared Rural Network where they will collectively invest £532mn, with the government investing £500mn, to address total not-spots and areas with poor coverage. Additionally, Vodafone, Hutchison and Telefónica have signed an agreement to address partial not-spots, sharing the new build between them (with Cornerstone adding sites for Vodafone and Telefónica).



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**Figure 38(a):** European MNO-towerco deals since 2007

Year	Country	Seller	Buyer	Tower count	Deal value (€)	Cost per tower (€)	Deal structure
2024	Bosnia and Herzegovina, Montenegro, Serbia	Telekom Srbija	Actis	1,800			SLB
2023	Malta	GO	BMIT Technologies	280	47100000	168,214	SLB
2023	Poland	Iliad	Cellnex		510000000		30% stake purchase from Iliad in OnTower JV
2023	Bulgaria, Croatia, Slovenia	United Group	TAWAL	4,800	1,220,000,000	254167	SLB
2022	Belgium	Telenet	Digital Bridge	3,322	745,000,000	224,262	SLB [1]
2021	Iceland	Sýn and Nova	ÍslandsTurnar	367	91,000,000	247956	SLB
2021	Portugal	MEO	Cellnex	223			SLB [2]
2021	Russia	MegaFon	Kismet Capital Group	17,000			Merger with Vertical [3]
2021	Russia	VEON	Service Telecom	15,400	847,800,000	55,052	SLB
2020	Portugal	Altice	Cellnex	223			SLB [4]
2021	Malta & Cyprus	Monaco Telecom	Phoenix Tower International	815			SLB
2021	Poland	Polkomtel	Cellnex	7,000	1,600,000,000		[5]
2020	Austria, Denmark, Ireland, Italy, Sweden, UK	Hutchison	Cellnex	24,600	10,000,000,000		SLB [6]
2020	Poland	Iliad	Cellnex	7,000	800,000,000	190,476	SLB with 40% equity
2020	Germany	Telefónica	Telxius	10,100	1,500,000,000	148,514	SLB
2020	Ireland	eir	Phoenix Tower International	650	300,000,000		SLB

Figure 38(a) continued: European MNO-towerco deals since 2007

Year	Country	Seller	Buyer	Tower count	Deal value (€)	Cost per tower (€)	Deal structure
2020	Portugal	Nos	Cellnex	2,000	375,000,000	187,500	
2019	Italy	Vodafone	INWIT	11,000	2,140,000,000		Merger INWIT & Vodafone Towers [7]
2019	UK	BT	Cellnex	220			Manage and market
2019	France	Iliad	Cellnex	5,700	1,400,000,000	245,614	SLB with 30% equity
2019	Italy	Iliad	Cellnex	2,200	600,000,000	272,727	SLB
2019	Switzerland	Salt	Cellnex	2,800	700,000,000	250,000	SLB with 10% equity [8]
2017	Italy	Wind (Veon)	Cellnex		77,000,000	104,054	SLB 10% stake out
2017	Spain	MasMovil	Cellnex	551	36,000,000	65,336	SLB
2017	Switzerland	Sunrise	Cellnex, Swiss Life and DTCP	2,339	430,000,000	183,839	SLB [9]
2017	France	Bouygues Telecom	Cellnex	600	170,000,000	283,333	SLB
2017	France	Bouygues Telecom	Cellnex	1800	500,000,000	277,778	SLB
2016	France	Bouygues Telecom	Cellnex	230	80,000,000	347,826	SLB
2016	France	Bouygues Telecom	Cellnex	270	67,000,000	248,148	SLB
2016	Spain	Telefónica	Telxius	11000			SLB
2016	Germany	Telefónica	Telxius	2350	587,000,000	249,787	SLB
2016	Ukraine	Lifecell	UkrTOWER	811	47,820,000	58,964	SLB
2015	Italy	Wind (Veon)	Cellnex	7,377	693,000,000	94,892	SLB with 10% equity [10]

Figure 38(a) continued: European MNO-towerco deals since 2007

Year	Country	Seller	Buyer	Tower count	Deal value (€)	Cost per tower (€)	Deal structure
2014	Spain	Telefónica/Yoigo	Cellnex	4,277	385,000,000	90,016	SLB
2012	France	Bouygues Telecom	FPS Towers	2,166	185,000,000	100,400	SLB with 15% equity
2012	Germany	KPN	American Tower	2,031	393,000,000	193,501	SLB
2012	Netherlands	KPN	Protelindo	261	75,000,000	287,356	SLB
2012	Netherlands	KPN	Shere Group	460	115,000,000	250,000	SLB
2012	Spain	Telefónica	Cellnex	500	45,000,000	90,000	SLB
2010	Netherlands	KPN	Open Tower Company	500			SLB
2008	Netherlands	KPN	Open Tower Company	101			SLB
2007	Ireland	Eircom	Towercom	338	155,000,000	399,988	SLB

[1] Site count includes 2,158 owned sites and 1,164 third-party sites

[2] Deal also includes 464 small cells

[3] MegaFon had previously carved out its assets into First Tower Company. First Tower Company is being acquired by Kismet Capital Group. Kismet Capital Group are also majority shareholders of Russian towerco Vertical which owns 5,000 sites. Vertical's sites will be combined with those of MegaFon under a new Kismet Capital Group owned towerco

[4] The deal also included 464 micro sites and had a total deal value of €209mn

[5] Acquisition also includes active equipment including c. 37,000 radio carriers, c. 11,000 km of fibre backbone and FTTT backhaul and a national network of microwave radiolinks

[6] Deal includes 8,900 sites in Italy, 1,150 in Ireland, 2,650 in Sweden, 1,400 in Denmark and 4,500 in Austria. In the UK, Cellnex acquired interests in or deriving from 6,000 sites. Deal value of €10bn consists of 8.6bn in cash and €1.4bn in shares in Cellnex, equivalent to a 5% stake in the company.

[7] INWIT paid €2.1bn to acquire 43% of Vodafone towers whilst Vodafone were issued €360mn INWIT shares to obtain a 37.5% stake in INWIT (equal to TIM's share)

[8] Cellnex has since acquired the remaining 30% stake that it did not own

[9] Cellnex has since acquired DTCP's stake

[10] Cellnex has since acquired the remaining 10% stake that it did not own

Figure 38(b): European towerco M&A since 2015

Year	Country	Entity/Seller	Buyer	Tower count	Deal value (EUR)	Cost per tower	Deal structure
2025	Serbia	Dot Networks	Connectis Tower	50			Portfolio acquisition
2024	Sweden	European Connectivity Networks AB	Telia Towers	10			Portfolio acquisition
2024	Ireland	Cellnex	Phoenix Tower International	1,900	971,000,000	511,000	Company acquisition
2023	UK	WHP Estates	Cellnex	55			Company acquisition
2023	Germany	NOVEC GmbH	Phoenix Tower International	220			Company acquisition
2023	Denmark & Sweden	Cellnex	Stonepeak		730,000,000		49% stake
2023	Poland	American Tower	Emitel	65			Portfolio acquisition
2022	France	Cellnex	Phoenix Tower	3,226			Portfolio acquisition [1]
2022	Russia	Russian Towers	New Towers	7,000			Company acquisition
2021	France	Hivory	Cellnex	10,500	5,200,000,000	495,238	Company acquisition
2021	Netherlands	T-Mobile Infra	Cellnex & DIV	3,150			Merger[1]
2021	Germany & Spain (plus Brazil, Peru, Chile & Argentina)	Telxius	American Tower	30,722	7,700,000,000	250,635	Company acquisition
2020	Italy	Towertel	Phoenix Tower International	1,000	NDC		Company acquisition [2]
2020	Portugal	OMTEL	Cellnex	3,000	800,000,000	266,666	Company acquisition
2019	UK	Arqiva	Cellnex	7400	2,300,000,000	310,811	Company acquisition[3]
2019	Ireland	Cignal	Cellnex	546	210,000,000	384,615	Company acquisition[4]
2019	UK	Spyder	Digital Colony				Company acquisition

Figure 38(b) continued: European towerco M&A since 2015

Year	Country	Entity/Seller	Buyer	Tower count	Deal value (EUR)	Cost per tower	Deal structure
2019	Finland	Digita	Digital Colony	556			Company acquisition[5]
2017	Netherlands	Mom and Pop	Cellnex	32	12,000,000	375,000	Company acquisition
2017	Russia	Link Development	Service Telecom	400			Company acquisition
2017	Netherlands	Alticom	Cellnex	30	133,000,000	443,333	Company acquisition
2016	France	ITAS TIM	TDF	420			Company acquisition
2016	Germany	WDR	American Tower				Company acquisition
2016	France	FPS Towers	American Tower	2,482	697,000,000	280,821	Company acquisition
2016	UK & Netherlands	Shere Group	Cellnex	1,004	393,000,000	391,434	Company acquisition
2016	Netherlands	Protelindo	Cellnex	261	109,000,000	417,624	Company acquisition
2015	Ireland	Coillte	Signal	113			Portfolio acquisition
2015	Italy	Tecnorad	EI Towers	134	17,000,000	126,866	Portfolio acquisition
2015	Italy	TowerCo	Cellnex	212	94,600,000	446,226	Company acquisition

[1] PTI will acquire 1,226 sites, whilst PTI with its JV partner, Bouygues Telecom will acquire 2,000 sites"

[2] Deutsche Telekom transferred T-Mobile Infra to a new investment fund, Digital Infrastructure Vehicle, receiving a €250mn cash payment and shareholding in DIV, equating to a capital commitment of €400mn in DIV. In parallel, Cellnex agreed to commit €200mn in capital to DIV. DIV then transferred T-Mobile Infra into Cellnex Netherlands, obtaining a 38% stake in Cellnex Netherlands. Cellnex retains a 62% stake in Cellnex Netherlands

[3] Deal also includes agreement for PTI to market 1,600 broadcast sites owned by EI Towers (Towertel's previous owners

[4] Deal also includes the right to market 900 additional Arqiva sites

[5] Signal own the land but not the tower on a proportion of sites. Cellnex will also inherit a pipeline of 600 new sites to be build by 2020

[6] Deal also includes data centres and fibre

Figure 38(c): Major European towerco equity deals and listings since 2016

Year	Seller	Entity	Tower count	Buyer/stock exchange	Equity	Deal value	Implied EV
2024	Turkcell	UkrTOWER and lifecell	1,200	DVL Telecom (NJJ Holding)	100%	\$524,300,000	
2024	Cellnex	Cellnex Austria	4,654	Vauban Infrastructure Partners, EDF Invest, MEAG, and ERGO.	100%	803,000,000	
2023	Denmark & Sweden	Cellnex		Stonepeak	730,000,000	4,600	49% stake
2023	Telenor Communication II AS and Participatiemaatschappij Vlaanderen NV	Norkring België NV	25	Cordiant Capital	100%		5,250,000
2023	Liberty Global	Cornerstone	15,464	GLIL Infrastructure	16.67%	360,000,000	
2023	Infrabridge	Towercom	409	John Laing Group	100%		
2022	Vodafone	Vantage Towers	45,900	KKR & GIP, PIF	81.7%	3,200,000,000	16,200,000,000
2022	Deutsche Telekom	GD Towers	40,500	Brookfield & DigitalBridge	51%	8,925,000,000	17,500,000,000
2022	Telia	Telia Towers Sweden	1,600	Brookfield & Alecta	49%	225,000,000	458,000,000
2021	Síminn	Míla	500	Ardian	100%	520,000,000	520,000,000 [1]
2021	PPF Group	CETIN	10,223	GIC	30%		
2021	Telia	Telia Towers Finland & Norway	3,700	Brookfield & Alecta	49%	722,000,000	1,473,460,000
2021	Macquarie	České Radiokomunikace	800	Cordiant Digital Infrastructure	100%		[2]
2021	American Tower	ATC Europe	25,274	Allianz & CDPQ	48%	2,600,000,000	5,416,667,000
2021	Vodafone	Vantage Towers	82,000	Frankfurt Stock Exchange	18.30%	2,200,000,000	12,021,858,000

Figure 38(c) continued: Major European towerco equity deals and listings since 2016

Year	Seller	Entity	Tower count	Buyer/stock exchange	Equity	Deal value	Implied EV
2020	AMP Capital	Axion	635	Asterion Industrial Partners	100%	NDC	-
2019	3i	Wireless Infrastructure Group	2,200	Brookfield			686,900,000
2018	Telefónica	Telxius	16,000	Pontegadea	10%	379,000,000	3790000000
2018	Altice	Hivory/SFR Towerco	10,198	KKR	40%	1,799,000,000	4497500000
2018	Altice	OMTEL/ Towers of Portugal	2,961	Morgan Stanley & Horizon Equity Partners	75%	495,000,000	660000000
2017	Telefónica	Telxius	16,000	KKR	40%	1,300,000,000	3250000000
2016	American Tower	American Tower Germany	2,197	PGGM [3]	49%	NDC	-
2016	Antin Infrastructure Partners	Axion	584	AMP Capital	100%	NDC	-
2016	New equity investment	Wireless Infrastructure Group	2,000	3i	NDC	300,000,000	-
2015	Telecom Italia	INWIT	11,200	Milano Italia Borsa	40%	875,300,000	2188250000
2015	Abertis	Cellnex Telecom	15,091	Bolsa da Madrid	66%	2,138,000,000	3239393939
2015	Coillte	Telecom assets	300	InfraVia Capital Partners	100%	70,000,000	70,000,000

[1] Mila's assets also include fixed line infrastructure, backhaul and active equipment

[2] České Radiokomunikace's tower portfolio includes broadcast as well as telecommunications site. The infraco also owns fibre and offers colocation and cloud services from its Prague data centre

[3] American Tower's German assets were transferred to ATC Europe in which PGGM has a 49% stake

